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Integration of traditional, transitioning & transformative digital technologies for value co-creation in B2B: A process model

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ABSTRACT

The nuanced understanding of process that firms in the B2B setting undergo for VCC and the role different types of technologies play in that process is missing. We adopted a research design that comprised of four sequential phases to address this gap. In first phase, we conducted a systematic review of literature. In second phase, we deductively developed an integrated 3 × 3 framework illustrating the usage of different types of technologies across the three stages of VCC. In third phase, we conducted semi-structured interviews with 25 practitioners to validate the framework. Finally, in fourth phase, we developed a process model to explain how digital technology can be leveraged in each step of VCC. The process model in the backdrop of integrated framework is novel as it codifies the individual activities of VCC and brings out the nuances involved at the interaction of the activities and stages. The findings can catalyse B2B VCC at a higher speed by establishing faster feedback loops through the integration of digital technologies in each of the process steps. The gaps in research and practice on how businesses co-creating value can iterate to get a testable product with a measurable outcome while leveraging digital technologies are discussed.

1. Introduction

The context in which businesses run today has changed drastically over the past few years. An industrial industry (also called as business-to-business or B2B; Kohtamaki & Rajala, 2016) is fraught with huge investments, longer design and delivery lead times, multiple stakeholder involvement, and heterogenous technological competencies and domain-expertise – all leading to a high uncertainty on the outcome. The impact of digitalization, accelerated with the advent of COVID, has been radical for the businesses and customers. The boundaries and market barriers are thinning due to digitalization and its impact on the interactions (Moravcikova, Krizanova, Kliestikova, & Rypakova, 2017; Ramaswamy & Ozcan, 2018). Suppliers and firms use the digital tools and technologies such as internet of things (IoT), social media, big data, analytics, artificial intelligence (AI), augmented and virtual reality (AR&VR), to name a few (Chang, Wang, & Arnett, 2018; Rindfleisch, O'Hern, & Sachdev, 2017) to come together and co-create value.

Value co-creation (VCC) in B2B is very different from well-researched B2C setting. B2C setting is characteristically different due to the source of value to customer, catering to the ever-evolving needs of the customer, and needs for flexibility to adopt product/service. Kostis and Ritala (2020) highlight this uncertainty in terms of *"interpretive uncertainty – the misalignment of views on the process and outcomes of co-creation due to differences among the involved actors (firms and individuals)"*. VCC in B2B setting involves high collaboration amongst stakeholders (such as firms, suppliers, system integrators, platform providers and industrial customers) to reduce the uncertainty and deliver outcomes to customers with confidence. Theories such as Service-Dominant Logic and Resource Based view, which had earlier dominated in B2C, is now being extended in the field of B2B (Hollebeek, 2019; Schulz, Zimmermann, Böhm, Gewald, & Krcmar, 2021). Recent literature studies in B2B setting have focussed on the open innovation

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culture that aids in the innovation with customers being part of VCC stages (Zhang & Xiao, 2020). A synergistic integration, that is, co-creating value through amalgamation amongst various stakeholder is much needed (Sarker, Sarker, Sahaym, & Bjørn-Andersen, 2012) and is widely achieved through adoption of digital technologies.

The usage of digital technology in the process of VCC is gaining significance. Increased connectivity, interdependence and co-evolution of actors and digital technologies facilitate relationships and networks (Aarikka-Stenroos & Ritala, 2017a; Hartmann, Wieland, & Vargo, 2018). Some of the recent literature illustrates how social media has an impact on B2B sales (Itani, Badrinarayanan, & Rangarajan, 2023; Sundström, Alm, Larsson, & Dahlin, 2021; Cartwright, Liu, & Raddats, 2021). The customers participate in VCC where they acquire, analyze and act on the big data (Zhang & Xiao, 2020) and take an active role as CDA (Customer as Data Analyst). Even though research has discretely documented the application and/or impact of digital technology on cocreation, the synthesis of this entire research within B2B setting would be of value for progressing the investigation from a dyadic relationship of buyer and supplier. Given this background, the first research question we aim to answer is:

RQ1. What are the different types of digital technologies relied on for implementing/achieving value co-creation (across different stages) in the B2B context?

Business run on processes that provide a definite start and end point with clear outputs. But we observe failures in businesses, due to lack of sustainable and consistent process, while there is a need to be agile to the changing needs. Process models are methods by which we design, explain, assess and optimize the process (Simutis, Oliveria, Manikowski, de Azevedo, & Lubber, 1997). B2B processes are more complex than B2C and more so during VCC, and hence there is a need for managers to design process models which will help them in process improvement and implementation of sustainable practices (Lindgreen, Di Benedetto, Brodie, & Jaakkola, 2021). The study by Grafmüller, Hankammer, Hönigsberg, and Wache (2018) illustrated the need for VCC, solution space development and information system design to overcome the challenges in collaboration between Small and Medium Enterprises (SME) for developing complex mass-customized products. Digital technology has advanced by leaps and bounds in the past few years, but digital technology usage by itself, without a process model, is neither efficient nor effective. Leveraging digital technology for VCC helps businesses have the right predictability and stability in their process and establishes the linkage to business outcomes. The existing literature in B2B VCC context do not bring out succinctly the process model and how practitioners/researchers can leverage digital technology effectively. Therefore, developing an integrated framework and process model by gathering empirical evidence shall help address this gap and provide inputs to researchers and practitioners on leveraging digital technology. This leads us to our second research question:

RQ2. How (the process) can firms in the B2B setting successfully integrate digital technologies during value co-creation?

To answer these two research questions, we adopted a research design that comprised of four sequential phases. In the first phase, we conducted a systematic review of the literature, which confirmed the gap identified and the synthesis informed the subsequent phases. The thorough review helped us to learn about the usage of digital technology in the process of VCC in B2B environment. The usage of the systematic approach helps ensure that our findings are comprehensive, unbiased and repeatable (Ferenhof & Fernandes, 2016; Palmatier, Houston, & Hulland, 2018). In the second phase, based on the descriptive and categorical analysis of literature, we deductively developed an integrated 3 × 3 framework illustrating the usage of different types of digital technologies across the stages of VCC. Content analysis proposed by Bardin (2011) was performed to further analyze the information as it is replicable and accurate (Krippendorff, 2018), offers a mechanism for document analysis that is systematic and rigorous (White & Marsh, 2006), and provides a concise summary through categorization (Elo & Kyngas, 2008). The analysis revealed an integrated 3 × 3 framework illustrating the usage of three broad types of digital technologies across the three stages of VCC. In the third phase, we conducted semi-structured interviews with 25 practitioners in the industry to validate the framework developed in the second phase. Finally, in the fourth phase, we develop a process model to answer how different types of digital technology can aid or be leveraged in each process step of VCC stages in B2B. The findings are expected to catalyse B2B VCC at a higher speed by establishing faster feedback loops through the integration of digital technologies in each of the process steps. The findings also explain the fit and impact of digital technologies for VCC in a B2B setting to remain competitive in the market.

The remainder of the paper is structured as follows. Section 2 details the concept of VCC, usage of digital technology and evolution of process models associated with VCC in B2B environment from existing literature. Section 3 describes the research methodology explaining each of the four sequential phases. Section 4 focusses on the synthesis of the systematic review, creation and validation of the 3 × 3 integrated framework, and finally the development of process model. Section 5 discusses the integrated framework and the usage of digital technology in each step of the process model. Key research and managerial implications, limitations and avenues for future research are also discussed in this section. Finally, Section 6 concludes the research.

2. Research background

Value is generally defined as the balance between benefits and sacrifices (e.g., price, cost) as perceived by the customer (Ravald & Grönroos, 1996). The benefits and sacrifices are customer-specific and is realized during the consumption or usage (Eggert & Ulaga, 2002; Vargo & Lusch, 2004a, 2004b). VCC is a process where the stakeholders (firms, suppliers and customers) get together to collaborate, brainstorm and ideate to create value for each other (Galvagno & Dalli, 2014; Gronroos, 2008; Kao, Yang, Wu, & Cheng, 2016). Research on VCC gained momentum post the evolution of service-dominant logic (SDL) (Vargo & Lusch, 2004a, 2004b). Kohtamaki and Rajala (2016) analyzed the literature on co-creation of value and coproduction of value proposition in the B2B service and solutions. Their study brought out the variety of definitions and meanings associated with VCC and presented the insights into theories and methods used. The study revealed that despite clarifications on SDL, there exists ambiguity on how researchers are using the concept. Several researchers have conducted systematic literature reviews on the topics of SDL, Product Service Systems, and VCC (Bonamigo, da Silva, da Silva, & Werner, 2022; Bonamigo, Dettmann, Frech, & Werner, 2020; Li, Kumar, Claes, & Found, 2020; Ranjan & Read, 2021; Saha, Goyal, & Jebarajakirthy, 2022; Vural, 2017). A key takeaway from these reviews is that, for the VCC to be successful, firms need to find ways to work together effectively with their suppliers and create value for each other and their customers. Given that VCC in a B2B setting comprises of complex systems that involve many different stakeholders and components, development of a systematic and process-driven approach can guide in its successful implementation.

Based on grounded theory, the systematic literature review (SLR) by Madanaguli, Parida, Sjödin, and Oghazi (2023) provided a thematic analysis consisting of 3 stages of VCC in digital industrial platforms namely – value creation, value delivery and value capture. Value creation involves what is offered to the customer in terms of products and services. In value delivery, the focus is on enhancing the digital platform architecture capabilities, orchestrating complementors and coordinating value delivery processes. In value capture, which is the most difficult part, designing platform revenue models, assessment of risk and scaling value capture are critical. While Madanaguli et al. (2023) focused on digital platforms and their relationship to the three stages of VCC, Kostis and Ritala (2020) suggested an integrative model for

defining three digital co-creation practices for industrial customers, namely Value Identification, Value Iteration and Value Realisation. The digital co-creation practices help the actors to redefine their roles based on their involvement with those representing end customer and firm taking a more active role and directing teams towards one common goal. Clarity on the role boundaries help reduce the uncertainty amongst the actors while improving co-creation outcomes, enhanced further by accumulated experiences of the actors in this process. Value identification occurs when the service provider using the digital artifact invites the supplier to aid in co-creating the solution. Value iteration is the process where value is assessed, compared and reformulated as an iterative step between service provider and suppliers. Value realisation is the process where the solution is implemented and the value is recognized by the customer and service provider, as intended. Leveraging these three digital co-creation practices provide opportunities for firms and suppliers to transform their practices, both within and across the business, to build on the digital technologies and remain successful. We rely on the three VCC stages defined by Kostis and Ritala (2020) for developing the integrated framework and process model.

In the following sub-sections, we highlight how B2B is different from B2C, the usage of digital technology during the process of VCC in B2B environment, and process models for VCC.

2.1. VCC in B2B environment

VCC in B2B environment is contrastingly different to that of the B2C environment (Hein et al., 2019). First, the actors involved vary significantly between them. In a B2C, the primary actors are firm and its customers; while in a B2B context, multiple actors are involved starting with the firm, its customers and suppliers (Breidbach & Maglio, 2016; Kohtamaki & Rajala, 2016), and non-traditional actors such as industry consultants or outside experts (Hartmann et al., 2018). Second, VCC processes and practices are different as the complexity of a B2B platform is often higher to that of a B2C platform. For example, Internet of Things (IoT) does not rely just on software developers for adding value, but elicits participation of sensor manufacturers, and software and application companies in an environment of non-homogeneity (e.g., machines, processes). Third, the users of the B2B business-critical platform are legal entities and not private individuals (Ferenhof, Bonamigo, Rosa, & Vieira, 2022). Fourth, the services of B2B are more complex compared to B2C. In an IoT context, device management, compatibility with sensors and machines, and communication protocols for industrial customers needs to be met by the supplier. Fifth and final, business customers are considerably of large value than single customer and expect customization of services, products and price structure. Roser, DeFillippi, and Samson (2013) stated that as the expectation of a buyer in B2B setting is strategic, suppliers develop solutions for specific firms as in automotive and software industries (Ceccagnoli, Forman, Huang, & Wu, 2012). The purchasing duration is typically longer due to complex buying decision and firms take a longer view of the entire ecosystem compared to B2C context (Dollinger, Lodge, & Coates, 2018). Studies have shown that being transparent and involving customers in the decision making, helped the firms in influencing value-informed pricing (Taghizadeh, Rahman, & Marimuthu, 2022). Acknowledging and accounting for these contrasting differences while approaching VCC investigation in the B2B setting is a prerequisite, thereby clarifies the invalidity of learnings from B2C research.

In a B2B setting, risk-sharing and building on trust provide opportunities for the involved two firms to be creative during their ideation process (Kuusisto & Riepula, 2011; Sales-Vivo, Gil-Saura, & Gallarza, 2020; Swink, 2006). The study by Cohen-Vernik, Pazgal, and Syam (2019) found that VCC helps generate new ideas and help the firms grow. Many B2B studies have focussed on how to improve innovation and enhance supply chain capabilities (Hoecht & Trott, 2006; Roser et al., 2013; Saha et al., 2022). It is also observed that the application of VCC varies by sector. VCC followed in a banking sector is very different from VCC in a telecommunication sector. The basis of value creation is "value-in-use", that is value buyer derives when a product meets their want. Value creation is a joint responsibility of all parties involved with the focus on experiential value that customers see. With service being the fundamental basis of exchange due to the application of the operant resources (knowledge and skill) for the benefit of another actor, it thus becomes critical to have right environment for VCC between the multiple stakeholders (customers, firms, suppliers). Based on Principal-Agent theory (moral hazard and information asymmetry), there exists a danger of shirking or over-compensation since the efforts from co-creation and the value derived by end buyer may not be visible to the supplier (Jaaskelainen, Kortelainen, & Hinkkanen, 2013). Hence, we can conclude that VCC in a B2B environment is vastly different from the VCC in a B2C environment.

2.2. Usage of digital technologies for VCC in B2B environment

Digital technology plays a key role in transformation of the business to drive improvements in operations and for new product creation (Boufim & Barka, 2021; Aslanova & Kulichkina, 2020). A survey of 3500 business executives, managers and analysts across the world indicated that the adoption of digital technology is nascent, despite the advantages that exist (Kane, Palmer, Phillips, Kiron, & Buckley, 2017). The study indicated five practices that companies can adopt to advance in the digital maturity curve namely (a) implementing systemic changes (highly digital organizations are organized around cross-functional teams), (b) having longer planning horizons to enable the company to adapt to changing digital environments, (c) scaling small digital experiments to achieve larger impact on the organization, (d) becoming talent magnets (employees are prone to stay in mature digital organizations) and (e) securing leaders with a vision to lead digital strategy and commit resources to the vision. Leveraging the digital co-creation practices outlined earlier (Value identification, value iteration and value realisation) with digital technologies goes a long way in terms of operational efficiencies and revenue growth.

The evolution of digital technology has also led to various technology maturity models. A study by O'Leary (2009) examined three related models by Gartner in terms of technology adoption, maturity curve and strategic technologies. The maturity levels of the digital technology vary over time and is categorized into Obsolete (rarely used technology), Legacy (not appropriate for new developments), mature mainstream (robust technology and not much evolution), early mainstream (proven technology), adolescent (maturing technology), emerging (pilots and deployments) and embryonic (in labs). The adoption category would thus vary based on the digital technology used, ranging from laggards to innovators. The usage of the three models also brought in aspects related to fit of different research approaches (example case studies or design science applications). Ryan, Fenton, Wasim, and Scarf (2020) explored and defined digital maturity events using the Industry 4.0 model, creating a definition of events and placing various digital technologies on a scale of digital maturity. They defined 4 events – Event 1.0 – "Basic" elementary digital technology engagement (use of widespread technologies such as websites, email, mobile devices and social media to communicate), Event 2.0 – "Emerging" digital technology assisted engagement (basic plus evidence of greater interaction with digital technologies), Event 3.0 – "Developing" digital technology enriched engagement (emerging plus digital technology important to run the business) and Event 4.0 – "Integrated" dedicated digital technology engagement (fully integrated systems, data-driven events, optimized communication and digitally managed). Technological revolution leading to digital transformation has reshaped the way firms co-create in B2B context today. Technology enabled VCC processes are complex interactions amongst the actors in the ecosystem. Breidbach and Maglio (2016) performed a multi-case study of Information and Communication Technologies (ICT) firms to study roles and frequency of interpersonal exchanges. Based on the tasks involved, defying the formal roles (e.g.,

project manager), they classified into eight roles that are taken up, namely independent tasks (expert, quality controller), joint tasks (enabler, performer, task allocator) and facilitation of tasks (facilitator, governor, conductor). The study by Leone, Schiavone, Appio, and Chiao (2021) illustrated how artificial intelligence enables and enhances VCC to provide better care and an improved patient journey, amongst the technology suppliers, healthcare firms and patients. Digital technology brings with it challenges and opportunities as highlighted in the study by Bonamigo and Frech (2020) for industry 4.0. The authors also propose a theoretical framework for smart industrial service systems. Business customers are involved in the activities of co-innovation with the firm in aspects related to operational performance, new service market performance (NSMP), new service development (NSD) and return of investment (ROI). In today's digital age, platforms have started to play a major role in B2B context (Hein et al., 2019). An example is the IKEA launching a digital platform called "Co-Create IKEA" encouraging customers to develop new offerings and providing a platform for IKEA's suppliers to co-create with IKEA (Saha et al., 2022). The different maturity models and digital technology adoption is thus aiding in the process of co-creation in an industrial context.

Leveraging the past research (Kane et al., 2017; O'Leary, 2009; Ryan et al., 2020), we classify the digital technology maturity into three categories (a) Traditional technologies: technology is dated and well established. This is instances of usage of e-mail, file sharing, teleconferencing, video conferencing, tele-showing etc. (b) Transitioning technologies: technology has been in use for some time in various B2B enterprises such as CRM, ERP products or platforms/interfaces. These technologies help B2B enterprises to scale their organizations (c) Transformative technologies: technology is nascent and came into existence in recent few years with digitalization such as social media, AI, Big Data, IoT, AR/VR. These technologies are changing the way businesses work together and co-create.

Given that VCC and digital technologies will co-exist, it is prudent for companies in the B2B setting to understand the investment challenges. Firms and suppliers get higher profits when they all co-create together, rather than when they try to create individually (Cohen-Vernik et al., 2019). The study by Liu, Wu, and Shi (2023) found that when crowdfunding initiators make money from their projects, they are more likely to use a crowdsourcing platform. Conversely, when initiators lose money, they are less likely to use a crowdsourcing platform. This is because initiators are motivated to use a crowdsourcing platform when they believe that it will be profitable for them. If an initiator has a successful project, they are more likely to use the platform again in the future. However, if an initiator has an unsuccessful project, they are less likely to use the platform again, as they may see it as a waste of time and resources. The study's findings are important for both crowdsourcing platforms and initiators. Platforms should focus on attracting and retaining initiators who are likely to be successful. Initiators, on the other hand, should carefully consider the costs and benefits of using a crowdsourcing platform before launching their project. Investment challenges thus need careful consideration in a B2B setting involving digital technologies usage for VCC.

In summary, from the review, we see that different types of digital technologies are used by interacting firms for VCC. A digital technology maturity curve with the continuously changing technology is a need of the hour for buyers and suppliers. While different digital technology is used in the process of VCC in B2B as highlighted in research, the impact of this usage on quality of interaction, quality of the service/product, time to market, frequent customer feedback and cost savings is lacking. A synthesis of how different types of digital technologies can be used during the process of VCC would help both researchers and practitioners in terms of product development and feedback, thus providing value to the customer. Customer's participation during the entire process of VCC in a B2B ecosystem, leveraging digital technology, would also help B2B firms and suppliers to innovate their offerings and provide value to customer. Hence a synthesis on usage of different types of digital technologies, based on the digital maturity curve, in the context of VCC in B2B would be helpful, which is addressed in this paper.

2.3. Process models in VCC

With the ever-changing digital maturity and technological innovations, we see rapid strides being taken in many industries to change their business process. B2B customers complete 57 % of their buying process before they contact a sales representative and perform 67 % of the buying tasks online (CEB Global, 2018; Gerard, 2014; Think with Google, 2013). Covid advanced the digital maturity of many firms and today retailers provide chatbots for buyers to order (Steward, Narus, Roehm, & Ritz, 2019). Businesses uses nudges, leveraging customer's earlier experience and data associated with it, to gain more traction and enable a smoother buying process. Business run on processes that provide a definite start and end point with clear outputs. But businesses need agility and hence this could lead to failures if the processes are not well-defined and agreed amongst the parties. Process models are methods by which we design, explain, assess and optimize the process (Simutis et al., 1997), and they help managers in process improvement and sustainable practices (Lindgreen et al., 2021).

The study by Steward et al. (2019) brought out the evolution of the process model, from 1956 until 2019, from a B2B buying process perspective. They uncovered 7 themes – transactions (steps in a single transaction), situations (still single transactions but includes process variations across situations), influences (integrative model and impact of behaviours), responses (how marketing can prompt different consequences), relationships (models on evolution of relationships, transactions are one type of the exchange), networks (all parties included in the model) and journeys (multiple customer touchpoints across multiple journeys). In the journey theme, the impact of digital technology on the buying process became a focus, with digital technologies used to map and model customer journey across touchpoints. Usage of the digital technology such as Big Data and commercial software such as Google Analytics, Microsoft Visio and IBM Journey Designer have made it easier to map, model and implement solutions to understand the purchasing behaviour. Shapiro, Rangan, & Sviokla, 1992, based on techniques of business process reengineering, advocated managers to chart the order management cycle and identify the roles responsible for completion of the task. The managers can thus look at mapping the customer activities and experiences during the process. The key insight that they suggested was having a manager for entire fulfilment process for oversight, since dealing with customers were not just the job of the sales and marketing personnel, but it was cutting across many other functions. The usage of process models helps businesses to clearly draw boundaries where they can collaborate with their partners, cut down the slack in the process and ensure good customer experience. It is in this context, and hence relevant to this paper, that we adopt a process model, for the leveraging of digital technology (leveraging journey maps) in the process of VCC in a B2B environment. This shall be in line with the micro-level model focus (on individual process steps and their immediate contexts) and procedural models (best practices to guide real-world situations) (Wynn & Clarkson, 2018).

There have been different process models suggested in VCC field. The study by Payne, Storbacka, Frow, and Knox (2009) was one of the first to bring out a conceptual model brand relationship experience in the context of VCC and SDL, to manage the customer experience. This process model was further extended by Durugbo and Pawar (2014) based on amalgamation of functions for customer involvement and technique selection as determined by the customer, supplier and encounter domains. The interesting study by Hein et al. (2019) brought out the tenets of how B2B platforms utilize VCC practices, using a multiple case-study in the context of IoT platforms. The paper brought out the facets of how platforms encourage supply side, demand side and connections between them by integration of complementary assets, readiness of the platform and servitization through application

development respectively. The paper concludes by illustrating a scalable infrastructure based on platforms which enable VCC. Grafmüller et al. (2018) brought out the challenges such as lack of financial resources and technical competencies prevent higher collaboration between SMEs developing complex, mass-customized products. The study, based on a case study on mass customization, illustrated the need for VCC, solution space development and information system design to overcome these challenges. The business process indicated in the study highlights how VCC is critical in the initial phases of customer contact, requirement analysis and building the value chain due to the customized offering. The firm needs a better understanding of the capabilities of the network companies in terms of solution space and information system design.

The research has looked at usage of process model in the contexts above – predominantly in a B2B sales journey, but a gap prevails in the holistic understanding of leveraging different types of digital technology during VCC in B2B context. With the evolving digital technology landscape, it thus becomes imperative for B2B firms to leverage technology during the process of VCC. Technology usage by itself, without a process model, will be less efficient and effective due to subsequent activity uncertainty and lack of understanding of dependencies. A process model of VCC in B2B context would help depict the activities/steps involved, leveraging different types of digital technology at each step, thereby showcasing the impact on quality of the offering, time to market and agility in taking customer feedback for further refinement of the offering apart from driving cost savings. The process model would also help establish the need for collaboration amongst the firm, its suppliers and customers. The absence of nuanced understanding on the process to be followed for integration of digital technologies during VCC in B2B context is one of the important gaps that is addressed in this paper.

3. Methodology

This study adopts a research design that comprised of four sequential phases to synthesise the relevant literature, develop and validate the framework, and finally arrive at the process model for leveraging digital technology during VCC in B2B environment involving buyer and supplier (Fig. 1). In Phase 1, we systematically reviewed the extant research and performed content analysis to synthesise research and determine the gaps. An integrated 3 × 3 framework was developed in Phase 2 covering digital technology maturity and VCC stages. We validated the framework in Phase 3 with 25 semi-structured interviews. Finally, in Phase 4, we constructed the process model based on the analysis of data gathered through 25 semi-structured interviews.

3.1. Phase 1 - SLR

In phase 1, we performed two stages of the SLR as illustrated in Fig. 2. In the first stage, the SLR was employed to provide an integrated view of the state-of-the-art research on digital technology usage for VCC in B2B. Content analysis was carried out as proposed by Bardin (2011). Detailed analysis was followed in the second stage, post a review of the scope covering aspects of reading and selection of the papers, collating information/data relevant to the impact of digital technology during process of VCC in an electronic spreadsheet. This step helped synthesise the information to provide a basis for the known and establishing the unknown, while documenting the same in each step (Ferenhof & Fernandes, 2016; Jesson, 2011).

To identify the literature, we defined the research questions, keywords, search strings and inclusion/exclusion criteria. We came up with a list of keywords for the scope of this study and used Boolean operators to combine the keywords into search strings, testing them in the process to ensure most relevant results were returned. As keywords, it was decided to use "industrial service", "service industry", "service industries", B2B, "business-to-business", "business to business" to include papers related to B2B. For the search on VCC, we used the keywords "co-creat*" and "cocreat*" (Bonamigo et al., 2020; Kohtamaki & Rajala, 2016; Saha et al., 2022). After multiple combinations, we used the following search string ("cocreat*"OR "co-creat*") AND ("industrial service" OR "service industry" OR "service industries" OR B2B OR "business-to-business" OR "business to business") in Title, Abstract or Author keywords.

The inclusion criteria were peer-reviewed papers in English from Web of Science database. Thus, grey literature such as reports and non-academic research, languages other than English, databases other than Web of Science were not accounted in this review. An electronic spreadsheet was created to capture the descriptive details such as name of authors, year of publication, title of the article, name of the journal, and research theme.

3.2. Phase 2 and phase 3 – Framework development and validation

In phase 2, based on the qualitative analysis of the selected articles, an integrated 3 × 3 framework is developed on digital technology maturity against the stages of VCC. The selected articles were read by two of the authors independently to perform clustering of the articles based on the digital technology maturity and VCC stages (Corbin & Strauss, 1990; Eisenhardt, 1989). The relevance of each article in terms of VCC and digital technology formed the basis for the two authors (working independently) to classify the articles in the grid. The two authors then agreed on the placement of the relevant article in the grid.

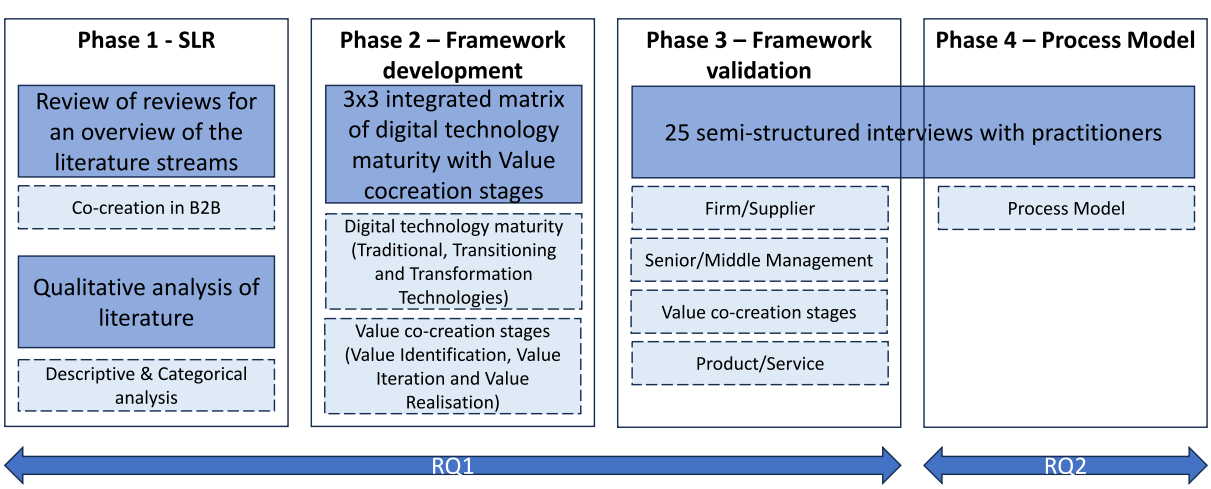

Fig. 1. Research design.

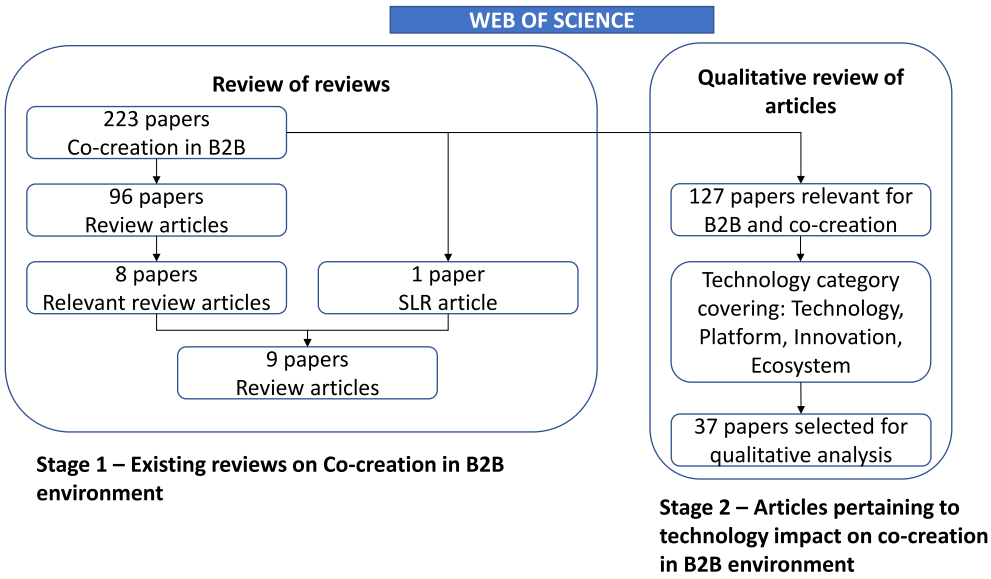

Fig. 2. Stages in Systematic Literature Review.

In the case of disagreements, they were resolved through discussion leading to the development of the integrated framework.

We conducted 25 semi-structured interviews to validate the 3 × 3 integrated framework for Phase 3 and arrive at the process model in Phase 4. We followed the interview process as outlined by Yin (2014). Interviews were conducted based on a defined set of questions (Appendix 1) and was semi-structured with flexibility to alter them to get more depth and detailed responses on usage of digital technology during VCC process (for Phase 4). Each interview was recorded and transcribed. Before starting the interview, we informed the participants that their identity would not be revealed during the research project, to assure the confidentiality of information shared. The other authors who performed the analysis were kept separate from the interview to avoid contamination of the data (Bernard, 2002). The interviewees were from a spectrum of digital technology maturity, sector, and roles as indicated in Table 1. All interviewees had work experience in B2B companies being part of the buyer, supplier or both entities, and with specific knowledge of VCC. To bring in different perspectives, different titles and roles were included, ranging from Senior Manager, Technical Manager, Delivery Manager, Delivery Partner, Head of Sales, Chief Delivery Officer to Chief Executive Officer. All the respondents had used traditional technologies and were now part of either transitioning technologies or transformative technologies, or both. The interviewees had used a mix of digital technologies including SAP/CRM, Open Source, Big Data, AI/ML and worked in platforms/traditional technologies. Hence, the interviewees had seen the spectrum of digital technology change and the adaptation of VCC as a process in their respective organizations. The interview time ranged from 30 min to 90 min with an average time of 48 min. Our protocol was aimed at eliciting information in the context of co-creation in B2B companies. The validation of the framework through these interviews helped in identifying gaps in existing literature and revealed areas for future research.

3.3. Phase 4 – Process model

Based on the transcripts from the 25 interviews, thematic analysis and selective coding was employed by two of the authors for arriving at the process model. Post completion of the interview process, and in line with the relevant literature, we performed coding, clustering, and reduction to obtain a coding scheme (Corbin & Strauss, 1990; Eisenhardt, 1989). The three steps in qualitative data analysis as outlined by Gioia, Corley, and Hamilton (2013) and Corley and Gioia (2004) was employed, leading to first order concepts, second-order themes and aggregate dimensions. Statements related to VCC in B2B environment and process steps are retained for analysis. The transcripts from the interviews are manually read by two of the authors independently. The two authors then agreed on the relevance of the statements and disagreements, if any, are resolved through discussion. Going through the stages of refining and data coding, the authors agreed on first order concepts (process steps). The two authors, working independently, labelled each statement for specific themes based on the order in which an output from one step flows to the next step. Three respondents had different viewpoints on whether the "solutions" should be in value identification or in value iteration stages. A second interview with these three respondents helped clarify their statements and arrive at appropriate position in the process model. Group of steps leading to an outcome for either the firm or supplier and leading to a logical decision point was used as the basis by the two authors to classify the second-order themes independently. The two authors then discussed the themes and the classification of statements to second-order themes. Any disagreement was resolved through discussions. When basic themes occurred frequently and led to logical outcomes from the process steps, second-level themes were identified. The ordering of the process steps for the second-level themes was independently performed by the two authors. Where there was any disagreement, this was resolved through discussions. Through selective coding (Corbin & Strauss, 1990), we compared the second order themes against each other to theorize the observed sequences and relationships amongst them. We summarize the data structure in Appendix 2 (Figs. 2.1-2.6). We superimposed these abstract categories resulting from previous coding over the business process steps and the demarcation of the processes across the three stages of VCC, namely Value Identification, Value Iteration and Value Realisation. This approach helped us to develop a sequential and actionable process model for using digital technology during VCC in B2B environment.

4. Findings

4.1. Phase 1 – SLR

4.1.1. Review on VCC in B2B environment

Before proceeding with the review on usage of digital technology for VCC, it is important to understand the literature reviews published on co-creation. Research papers published until October 2023 were

Table 1
Semi-structured interview - respondents profile.

Particulars	Count	Percentage	Particulars	Count	Percentage
Worked in Firm/Supplier/Both			*Respondent Designation*		
Firm	5	20.00 %	CEO (CEO-1, CEO-2)	2	8 %
Supplier	10	40.00 %	Chief Delivery Officer (CDO)	1	4 %
Both	10	40.00 %	Chief Marketing Officer (CMO)	1	4 %
Grand Total	25		Chief Talent Officer (CTO)	1	4 %
Respondent Experience			Chief Information Officer (CIO)	1	4 %
Middle Management	15	60.00 %	Data Scientist (DS)	1	4 %
Senior Management	10	40.00 %	Delivery Head (DH)	1	4 %
			Delivery Manager (DM-1, DM-2)	2	8 %
Value Co-creation Life cycle			Delivery Partner (DP-1, DP2-, DP-3, DP-4, DP-5)	5	20 %
Value Iteration, Value Realisation	8	32.00 %	Delivery Partner & Entrepreneur (DP&E)	1	4 %
Value Realisation	10	40.00 %	Head of Consulting (earlier Global Head of Delivery) (HoC)	1	4 %
Across the 3 stages	7	28.00 %	Head of PMO (PMO)	1	4 %
			Head of Sales (HoS)	1	4 %
Digital Technology Maturity - Awareness and Usage			HR Manager (HR)	1	4 %
Traditional Technologies	NA		Management Consultant (MC-1, MC-2)	2	8 %
Transitioning Technologies	9	36.00 %	Marketing Manager (MM)	1	4 %
Transformative Technologies	13	52.00 %	Technical Architect (TA)	1	4 %
All Technologies	3	12.00 %	Senior Manager (SM)	1	4 %
Sectors			*Years of VCC Experience*		
IT	14	56.00 %	5–10 years	9	36 %
Manufacturing, IT	3	12.00 %	10–15 years	12	48 %
Pharma	2	8.00 %	15 years and above	4	16 %
Oil & Gas, IT	2	8.00 %			
Education, IT	1	4.00 %			
Retail, IT	1	4.00 %			
Manufacturing, Defence	1	4.00 %			
Retail	1	4.00 %			

gathered from Web of Science using the search criteria and this yielded 223 records. The number of papers with type as "Review" led to 96 papers. We then individually examined the title, abstracts and keywords and the relevant review papers came down to 8. We reviewed the remaining papers and found 1 more article which was a SLR but had the manuscript type as "Article". These nine papers were then critically analyzed, and the details of the analysis are as stated in Appendix 3. These reviews on VCC synthesised actors involved, theoretical lens used and constructs for VCC, and opportunities and challenges in IoT in the industrial services context. As seen in Appendix 3, there exists no review on digital technology usage for VCC.

4.1.2. Qualitative review of articles

In the second stage of the SLR process, we reviewed the title, abstract and author keywords individually and found 57 papers that are not relevant for the objectives of this study. The remaining 127 papers were then categorized into various research themes and 37 papers were found relevant from usage of digital technology on VCC in B2B environment. After going through each of the papers in detail, we found 37 papers relevant from a digital technology usage perspective. These 37 papers were then fully analyzed for the research objectives. The attributes for the qualitative analysis were industry, methodology (quantitative, qualitative, or mixed method), specific qualitative approach or quantitative analysis technique, digital technology used, VCC stages (amongst the three VCC stages), theoretical lens, and themes emerging against the VCC stages. It is important to know which digital technology was being used and in what life cycle of the VCC stages, and hence the parameters related to technology usage and VCC stage is important (specifically to answer RQ1).

Digital technology usage: Multiple technologies have been cited in the reviewed papers as seen in Table 2. Social media presence has been higher considering that sales personnel have used this medium for VCC activities with the customer. The usage of CRM is now aiding the sales personnel in the relationship management with customers in this domain. The next big usage is on big data, considering the large volumes of data that exist today. There are 3 reviewed papers which do not specifically mention the technology and is generic with respect to digital technologies. 10 of the 37 papers do not cite any specific digital technology usage. The advent of Industry 4.0 (including IoT) and usage of open innovation platform in conjunction with Robotics, Autonomous vehicles, and cloud computing are trends seen from the type of digital technology being used.

VCC stages: The integrative model suggested by Kostis and Ritala (2020) has been adopted to categorize the reviewed journals to the three VCC stages for industrial customers, namely Value Identification, Value Iteration and Value Realisation. Value identification occurs when the service provider using the digital artifact invites the supplier to aid in co-creating the solution. Value iteration is the process where value is assessed, compared and reformulated as an iterative step between service provider and suppliers. Value realisation is the process where the solution is implemented and the value is recognized by the customer and service provider, as intended.

Based on the reviewed papers, a temporal view of the VCC stages is

Table 2
Digital Technology spread.

Type of digital technology	Total
AI	2
Big data	4
CRM and Social Media	1
Digital	3
ERP	2
Industry 4.0	2
Metaverse	1
Mobile app	1
Not mentioned	10
Open Innovation Platform	1
Robotics, autonomous vehicles, open system integration, IoT and Big Data Analytics, cyber-physical systems, cloud-computing, augmented and virtual reality simulation, and cyber-security	1
Smart glasses	1
Social media (video clips, chat, blogs, presentations, LinkedIn, Facebook)	6
Video conferencing, telephoning, teleconferencing, email or file sharing/ intranet	1
Virtual Reality (VR)	1
Total	**37**

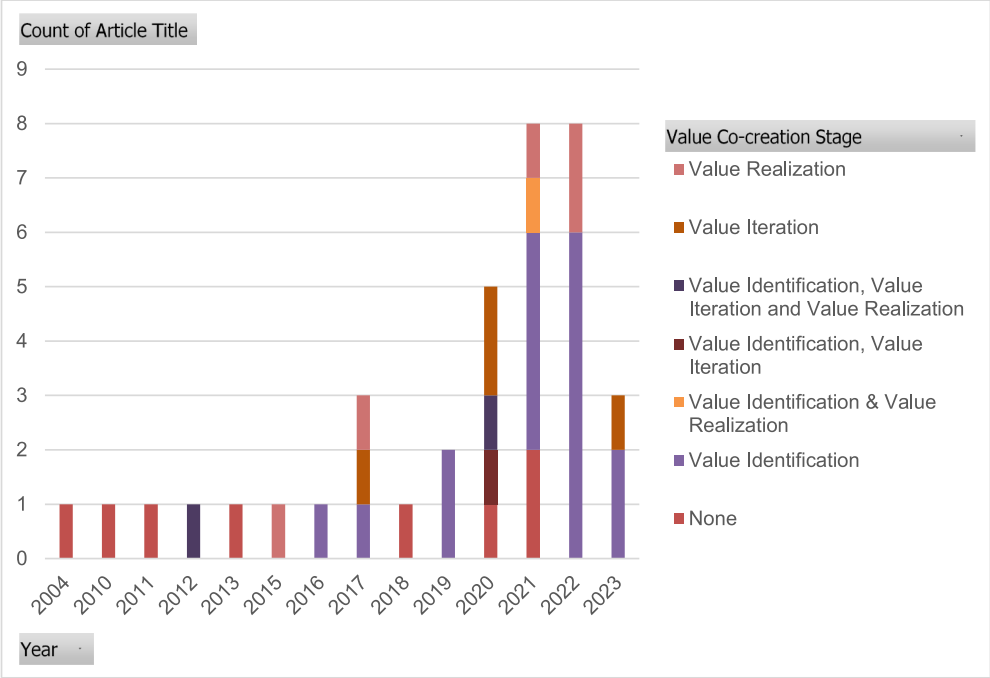

Fig. 3. Temporal view of VCC stages.

illustrated in Fig. 3. In the past 3 years, more articles are being published on value iteration and value realisation. As seen in Fig. 4, of the remaining reviewed papers, it is seen that value identification constitutes about 48 % and hence this stage is comparatively maturing. 21 % of the reviewed papers do not address the VCC stages and there is a need for value iteration and value realisation stages to evolve.

4.1.3. Themes

Based on the digital technology usage across the VCC stages, various themes emerging from the detailed analysis of 37 papers were determined to answer RQ2. For the reviewed journals, a clustering of the literature was done in line with VCC stages as stated by Kostis and Ritala (2020). The methodology of categorization is based on the analysis of the objectives and issues addressed in the papers reviewed. Appendix 4 covers the detailed descriptive and categorical analysis of the 37 papers. Table 3 summarizes the literature under the three VCC stages. Very few studies in value iteration and value realisation indicates that there is scope for further research to contribute to literature to derive theoretical and managerial implications.

4.2. Framework development

Fig. 5 illustrates an integrated framework of the peer-reviewed journals on the digital technology maturity with VCC stages. It is observed that the usage of digital technology in value iteration and value realisation is limited. Also, while new digital technology aspects such as social media, AI, Smart Glasses/VR, and platform/interfaces are being analyzed in the value identification phases, the relevance and the measurement of the same across the value iteration and value realisation stages of VCC needs further exploration and research. Looking at this framework from a domain perspective, it is seen that the peer-reviewed journals have touched upon agriculture area to show tangible outcome (Jayashankar, Johnston, Nilakanta, & Burres, 2020). Outside this, the number of peer-reviewed journals indicating the usage of either transitioning or transformative technologies are very limited. Hence, the full impact of the digital technology in VCC in B2B environment is yet to be seen.

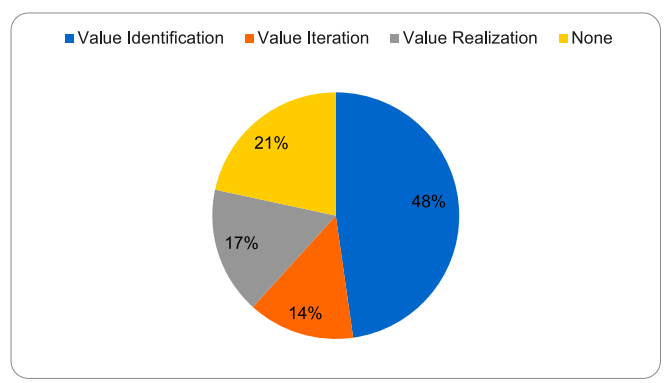

Fig. 4. Distribution of co-creation stages from peer-reviewed papers.

4.3. Framework validation

We present our findings for the framework validation based on data deduction and the analysis of 25 interviews. As we progressed during the interview analysis, we also started to cull out themes emerging from the different interviews. We support the validity of the framework based on representative evidence from our data in Table 4. The confirmation of the framework was a good start point for us to understand the process model for VCC in B2B as used across the various domains and digital technologies.

4.4. Process model

The three stages of VCC – value identification, value iteration and value realisation – formed the basis for understanding the steps in each of the stages and the flow of activities from one step to another. Table 5 indicates the addition, change and reorder to the process model based on the interviews conducted. An addition led to inclusion of a process step based on interviews, such as inclusion of “Partner value pitch” step in the business value articulation step of process model in value identification stage. A change to a process step led to additional meaning acquired based on interviews, such as the importance of “Solutions” step in

Table 3
Clustering of literature into co-creation stages.

VCC Stages	Learning	Author Name & Publication Year
Value Identification	The **roles** played by economic actors in technology-enabled VCC process	(Breidbach & Maglio, 2016)
Value Identification	The ecosystem approach for increasing interdependence and co-evolution of contemporary business and innovation activities	(Aarikka-Stenroos & Ritala, 2017b)
Value Identification	Social media and platforms for engagement based on SDL/ RBV-informed model	(Hollebeek, 2019)
Value Identification	Social innovation through VCC in context of B2B relationship between platform retailer and the manufacturer	(Zhang et al., 2021)
Value Identification	Companies shifting B2B sales towards value-based selling using a more proactive, continuous process wherein digital VCC plays a big role.	(Alamäki & Korpela, 2021)
Value Identification	Effects of customer-perceived value and trust	(Sundström et al., 2021)
Value Identification	Adopting the S–D perspective to analyze the link between value formation in B2C and B2B relationships in service ecosystem	(Schulz et al., 2021)
Value Identification	Role of customer emotions in embracing mechanics when participating in gamified platforms	(García-Magro, Martín-Peña, & Sánchez-López, 2023)
Value Identification	IoT engagement needs to be considered from three perspectives - multidimensional perspective (cognitive, emotional and behavioural), beyond a dyad perspective and importance of co-creation	(Soltani, 2021)
Value Identification	Integrative framework amongst firm's customer agility, new product success and capabilities with big data analytics	(Shirazi, Tseng, Adegbite, Hajli, & Rouhani, 2022)
Value Identification	Social media usage by salesperson critical to generate social capital	(Itani et al., 2023)
Value Identification	Salesperson's job autonomy and sales quota has a significant moderating effect on knowledge and VCC	(Itani et al., 2023)
Value Identification	How resources are combined in solutions based on AI, how suppliers and clients co-operate, and how outcome of the VCC can be measured for advanced technologies in B2B context	(Kot & Leszczyński, 2022)
Value Identification	A conceptual framework on impact of metaverse on creation and delivery of services experiences	(Gursoy, Lu, Nunkoo, & Deng, 2023)
Value Identification	How co-creation amongst lead firm and its contributors could lead to value destruction	(Sahaym, Vithayathil, Sarker, Sarker, & Bjørn-Andersen, 2023)
Value Identification, Value Iteration	In a platformized industry, transparency is contingent primarily on the nature of the market	(Bourne, 2020)
Value Identification, Value Iteration and Value Realisation	VCC in B2B alliances associated with selling, extending and implementing packaged software, especially ERP systems	(Sarker et al., 2012)
Value Identification, Value Iteration and Value Realisation	Adoption of VR technology and utilization of digital artifacts for implementing custom solutions in robotics/ automation projects, thereby increasing engagement, reducing uncertainty and improving project outcomes	(Kostis & Ritala, 2020)
Value Identification, Value Realisation	Multiple solution of AI supporting firms in co-operating value in B2B industrial markets	(Leone et al., 2021)
Value Iteration	Emergence of digital technologies significantly contributing to OM in manufacturing and service industries	(Agrifoglio, Cannavale, Laurenza, & Metallo, 2017)
Value Iteration	Customer co-creation in the operational agility process enables customers to contribute their complementary skills and knowledge thereby improving the firm's flexibility in adjusting to market changes	(Chuang, 2020)
Value Iteration	The active role played by customers in form of Customer as Data Analyst (CDA) in big data analytics	(Zhang & Xiao, 2020)
Value Iteration	Impact of social media within business relationships and networks	(McGrath, O'Toole, & Drummond, 2023)
Value Realisation	The relative influence of different business stakeholders in the context of the business modelling of two-sided Internet platforms	(Muzellec, Ronteau, & Lambkin, 2015)
Value Realisation	Digitalization capabilities (intelligence capability, connect capability and analytic capability) enable VCC with customers through perceptive and responsive mechanisms	(Lenka, Parida, & Wincent, 2017)
Value Realisation	Leveraging big-data technology, farmers co-create and realize value in monetary and non-monetary aspects	(Jayashankar et al., 2020)
Value Realisation	Theoretical contribution in bridging marketing, technology and experience literature with AI	(Neuhofer, Magnus, & Celuch, 2021)
Value Realisation	Capability of the company in digitalization, high customer satisfaction in existing relationship, building trust in IoT helps drive perceived value of IoT-based business models in B2B settings	(Falkenreck & Wagner, 2022)
Value Realisation	Integration of digital platforms and out-bound Open Innovation enables firms to establish vibrant innovation ecosystems	(Tsai, Mortara, & Ahn, 2022)

the business value articulation step of process model in value identification stage. A reorder of a process step was needed to bring coherence to the process flow based on interviews, such as "Solutions and partner validation" steps being brought up before "Technology Considerations" step in the business value articulation step of process model in value identification stage. In the following paragraphs, we discuss each of the

Digital Technology Maturity

	Value Identification	Value Iteration	Value Realization
Transformative Technologies	**AI/IoT/VR/Smart Glasses:** Enhance user's knowledge for better care and improved patient journey (Leone et al., 2021); Emotional engagement lowers the risk of disengagement (Soltani, 2021); Knowledge Acquisition for market and technology deployment (Aarikka-Stenroos & Ritala, 2017b); Design reviews to establish clarity through preference visualization, expectations comparison and integration of views; inputs co-creation, iterate with customers; solution optimization (Kostis & Ritala, 2020); ARA based framework to study co-creation in AI-activated value with focus on conversational agents (Kot & Leszczyński, 2022) **Metaverse:** Conceptual framework on impact of Metaverse on Service industries (Gursoy et al., 2023) **Social Media:** Business customer engagement as antecedent to customer co-creation using social media (Hollebeek, 2019); Action oriented messages have significant impact than product-oriented messages and value-based messages - leading to trust in company (Sundström et al., 2021); Effect of customer co-creation on operational agility using social media; directly influencing customer-firm relationships (Chuang, 2019); Enhance market knowledge and reputation using social media, which is then used to leverage networking skills (Itani et al., 2022) **Big Data:** Customer need tacitness positively moderates relationship between Customer as Data Analyst (CDA) and new product performance, and negatively moderates relationship between Customer as Data Provider (CDP) and new product performance (H. Zhang & Xiao, 2020); Analyze and access data (Aarikka-Stenroos et al., 2017); Connect marketers and organizations - bring transparency, Digital transformation journey to consolidate existing data, Collect new data and strengthen relationship with users, Open Banking (direct payments by clients) (Bourne, 2020); Firm's agility enhanced by big data aggregation and analytical tools, leading to product innovation (Shirazi et al., 2022)	**IoT:** Digitalization capabilities (intelligence capability, connect capability and analytics capability) enhance value co-creation in implementing a servitization strategy (Lenka et al., 2017); Customization of product/service has deep implications for operations management - customize onboard operations (Agrifoglio et al., 2017); Joint venturing between manufacturers and customers - buyer experience and buyer participation in these concepts increase a (pre-defined or newly developed) value in exchange for both parties (Falkenreck & Wagner, 2022) **Social Media:** Impact of social media within business relationships and networks. For example: Social media as a collaborative medium among B2B network actors (McGrath et al., 2023)	**Big Data:** Insights to improve farmer's financial sustainability and operations; Deepen farmer's knowledge about crop management; Exchange ideas with fellow farmers (Jayashankar et al., 2020) **Impact of AI** as an operant resource on event experiences. AI can co-create value for customers by recommending ways to save money and enhance experience flows, while realizing value for event organizers by creating faster processes and cost savings (Neuhofer et al., 2021)
Transitioning Technologies	**CRM:** Social media and CRM technology support sales reps in co-creating value with customers through knowledge brokering (Itani et al., 2023) **Platforms:** Synergistic integration and joint innovation; Simplicity and adaptability of technology (Sarker et al., 2012); Connecting ecosystem actors (Aarikka-Stenroos et al., 2017); Hyper-connected world and improved consumers' information availability in advertising (L. Zhang et al., 2021); Value based selling activities through deepening customer interactions, each actor in the ecosystem adding value (Alamäki & Korpela, 2021); Customers' emotional mechanics when participating in gamified platforms (García-Magro et al., 2023) **Interfaces:** Resource integration and service exchange in tripartite relationship in the use of service mobility apps (Schulz et al., 2021)	**Platform:** Business models of internet ventures evolve over time in terms of audience orientation and its business model (Muzellec et al., 2015); Orchestration of B2B platform more complex than B2C. Business users highly value the quality of platform offerings; market dynamics is quality driven. Integration of digital platforms and out-bound Open Innovation enables firms to establish vibrant innovation ecosystems (Chung et al., 2021)	
Traditional Technologies	**Email/Teleconferences/tele showing/Video Conferencing:** Facilitate resource exchange or resource generation (Breidbach & Maglio, 2016)		

Value Co-creation Stages

Fig. 5. Integrated framework illustrating digital technology maturity in VCC stages.

process steps for VCC stages and provide representative quotations (Tables 6–9) from the data to illustrate the same.

4.4.1. *Value identification*

During the value identification, the major processes would involve identification of the gap based on market research, opinion of experts, and SWOT (Strengths, Weakness, Opportunities and Threats) for products/services in alignment with firm's strategies. This would be followed by articulation of the business value for the gap identified based on value positioning, determination of market potential, leveraging the expert opinion, the projected return on investment (ROI), SWOT for talent/technology/operating model and business model (ways of working between the firm/supplier) and the risks involved. Suppliers could be involved with the firm in this aspect of business value articulation. Hence at this stage, a determination of which areas to co-create is firmed up since this is needed as part of the value positioning and projected ROI. Thereafter, the firm and supplier work together to gain alignment across solutions, partnership value and validation of partner, alignment on who owns the Intellectual Property (IP) for the co-created product, re-validation of pricing and ROI, and digital technology considerations. The firm and the supplier by the end of value identification are thus aligned and have a clear strategy for the next stages of VCC process.

4.4.2. *Value iteration*

In the value iteration, the firm and supplier co-create by way of ideation/brainstorming the solutions, leveraging proof of concepts/rapid prototype/framework/accelerators to get to a testable prototype. This prototype is then taken to a set of beta customers (with marketing effort leveraging digital technology) to measure the value seen by the customer, perform ROI validation and adapt the product/service based on the learnings. This is an iterative process.

4.4.3. *Value realisation*

In the final stage of Value Realisation, the product/service is sold to the end customers (with marketing effort) and value measurements are made to ensure ROI is validated and product/service can be adapted based on the customer's feedback and learnings during value realisation.

4.4.4. *Preconditions for VCC stages*

During the analysis of the data, we also realized that there are a few preconditions (Table 9) that needs understanding between the firm and supplier.

The absence of these preconditions is likely to have detrimental impact on the VCC outcome and ensuring a longer mutually beneficial relationship between the firm and the supplier. The process model is based on trust amongst the partners (firm and supplier), considerations for compliance and regulatory, supply chain contract management and governance processes.

4.4.5. *Process model*

The data analysis and the representative quotations for each of the VCC stages – value identification (to identify the gaps and opportunities for the firm), value iteration (to get a testable prototype and validate the ROI) and value realisation (product sold and value realized for the firm/supplier) – help us arrive at a process model for digital technology usage in B2B environment as illustrated in Fig. 6.

The outcome of the value identification is to identify a gap or an opportunity while leads to creation of a business plan, understanding the strengths of the organization and how they can leverage partners to co-create the product/service. One of the key steps in this stage is the ability to having winning propositions for both the firm and supplier, which ensures longer term relationship and creativity right through the partnership. IP protection and alignment is another key aspect that the firm and supplier would need to mutually agree to ensure there are no heartburns at a later stage to either party. In the process of the business articulation and viability of the product/service, the digital technology comes in handy to determine how quickly can we take the product/service to the market with quality.

The outcome of the value identification leads into the value iteration where the focus for the businesses would be to get a testable product/service which the end customers could use and provide invaluable feedback. This process could be aided by frameworks/accelerators,

Table 4
Framework validation.

Representative quotations	Themes
"the 3 × 3 framework looks appropriate covering the full life cycle and is holistic - in terms of co-creation stages and digital maturity" (CEO-1) *"matrix looks ok" (CTO)* "X and Y axis makes sense" (MM)	Framework validation
"Value identification is the easy in IT space. Value iteration/measurement/realisation has been difficult (a) convincing customer to adopt based on another customer even if contextualized (b) Budgets not marked for this year (or even in next 6 months)" (CEO-2) *"Co-creation is not easy - the systems are too complex, not straightforward. There are no products available in the market - due to complexity, processes not being well streamlined. The firm is moving from largest SAP journey to H/4 journey, but suppliers do not have practical experience for our context and hence hard to co-create with my firm" (DM-1)* *Intricacies for adoption of technology "In Transitioning technology, not everyone is familiar with products and hence need for simpler UI or make lighter apps to overcome internet bandwidth issues" (DP-3)* *"Traditional technology is easy and low risk; Transitioning technology, we are more confident, extension and adoption is easier. But we struggle in Transformative technology since the market and technology fit are unknown - adoption rate is very critical and known for failures, which is where we wish to co-create" (MM; DP-5)*	VCC stages and adoption
"Relook at traditional and transitioning technologies - are they getting merged today in the context of work we do today" (DP-5) *"AI can fit in any of the 9 blocks" (MM)* *"Composable architecture by domain is likely to happen in future. For example: Domains we operate in can become independent for each service provider – such as HIPAA compliance as a service through service provider (domain specific)" (DP&E)* *"Digital maturity varies based on market maturity" (MM)* *"Domain specific cloud and Storage by industry" (CDO)*	Digital Technology Maturity Curve
"Shiny stuff is in slides/conferences but does not help that much. We still continue to use Java, .net and Endure (they remain bread and butter); while we use Azure and PowerBI. Not so much on IoT & AI (not that we are against new technology) - all decisions based on Return on Investment" (DM-1) *"Automation is the buzzword - there are technologies that are proven, there are some that are yet to be proven" (DP-5)*	Challenges in Transformative Technologies
"More than technology, it is the value of what we deliver based on co-creation that is critical. Technology is an enabler. Hence third axis could be value in relation to co-creation stages" (TA) *"Role of Governance in the whole context of technology and co-creation" (CEO-1)*	Potential areas for research

rapid prototypes, high-fidelity designs through brainstorming sessions. This is also the stage where the business would get a view of whether the product/service sells in the market or would they need to persevere, pivot (change direction) or stop (cut losses). The value realisation stage ensures that the ROI that businesses had intended to achieve jointly has been achieved, with an evolving product based on learnings from end customer.

As indicated in the process model, the process remains iterative. At the value iteration stage, the business may understand that the core problem for the end customer is not resolved, and they may need to ensure the right gap is addressed, by going back to the value identification stage. The same could true even in the value realisation stage when it could lead back to either value identification or value iteration stage.

5. Discussion

We discuss in this section on how digital technology can be leveraged in each of the steps in the process model, articulating the importance of the 3 × 3 integrated framework and process model as a way forward, when the firm and supplier get together to co-create in a B2B context. The outcome of the extensive interviews helps us to synthesise future research directions, while clearly highlighting the implications and limitations of the study.

5.1. Leveraging digital technology to aid VCC

The study performed in this article elevates the discussion in Grafmüller et al. (2018) by recognizing that leveraging digital technology is a need of the hour in today's context. While the study by Grafmüller et al. (2018) was generic focussing on the relevance of VCC, this article extends that research by discussing the nuances at the intersection of VCC and digital technologies. Identifying the process steps in each of the VCC stages, as defined in Fig. 6, provides clarity and direction for firms and suppliers to collaborate. In the value identification stage, based on company strategy, market research, expert opinion and SWOT of their products/services, firms are equipped to identify a gap. Gap identification leads to process of determining the right valued product/service for the firm, performing a capability assessment of their talent, technology and operating model (Durugbo & Pawar, 2014; Gronroos, 2011; Payne, Storbacka, & Frow, 2008). It is at this stage the firm shall need to decide on whether it is best to close this gap in-house or should they leverage skilled suppliers. The partner value pitch then leads to greater alignment amongst the firm and supplier (Hein et al., 2019) with key considerations on technology and IP protection for the co-creation alignment.

The value iteration stage is iterative but provides opportunities for firms and suppliers to co-create together by way of ideation/brainstorming sessions, leveraging the strengths from each other to develop a rapid prototype, proof of concept or frameworks/accelerators (Durugbo & Pawar, 2014; Payne et al., 2008). The testable prototype is validated with the select end-customer (called Beta customers) and a determination of the value for the product/service as seen by end-customer is recorded (Durugbo & Pawar, 2014). This helps provide the right justification for the spend on product/service and firm and supplier can get back to drawing board to make amends to the product/service due to feedback in the Beta testing of Value iteration stage. In the final stage of Value Realisation, the product/service is in a good shape for mass customization and gets rolled out into the market with the help of marketing team with value as seen by end-customer recorded. The litmus test on the productionised product helps in ensuring ROI is validated (Payne et al., 2008), lessons learnt are further adopted to make better product. The entire process is iterative and provides opportunities to leverage digital technology through the spectrum of process steps in each VCC stage.

To aid the stages of VCC, it is critical to understand what type of digital technology can be used in each stage (as defined by Kostis and Ritala (2020)) and what benefits are expected to be derived. Table 10 illustrates the cross-section of the VCC stages and digital technology maturity on the process model. Based on the interviews conducted for Phase 4, this table links the VCC stages (Value Identification, Value Iteration and Value Realisation), related process steps (as brought out using the first order concepts, second-order themes and aggregate dimensions in the process model) to the usage of digital technology and

Table 5
Interviews to definition of Process Model.

Value Co-creation Stage	Process Model	Step	Addition/ Change/ Reorder	Representative quote from interview	Respondent Designation
Value Identification	Identify a gap	SWOT of products/ services	Change	*"Solutioning happens during value identification, where we identify the potential bottlenecks and how do we get them resolved"*	DP&E
Value Identification	Business Value Articulation	Market Potential & Expert Opinion	Addition	*"Company strategy is one level up of expert opinion? Consider 'market potential' for customer need"*	DP&E
Value Identification	Business Value Articulation	SWOT (Talent, technology, operating model)	Addition	*"SWOT (Talent, technology and operating model) part of business value articulation since ability of the firm to deliver value needs to be considered for value positioning"*	DP&E
Value Identification	Business Value Articulation	Cost Benefit Analysis, Feasibility Study	Addition	*"During business value articulation, we have considerations for analyzing the cost benefit and perform feasibility studies"*	MC-1
Value Identification	Business Value Articulation	Partner value pitch	Addition	*"Need a business model (more than just governance) - how does it work/engagement model ('what is in it for me' for both parties) and why should customer do it? Why should we work together, more so in a connected world"*	DP-5
Value Identification	Business Value Articulation	Partner value pitch	Addition	*"How will partner see value - what is the value pitch for partner in "should we co-create""*	DP&E, DP-4
Value Identification	Business Value Articulation	Partner value pitch	Addition	*"Interlock with partners required during value identification"*	DM-2
Value Identification	Business Value Articulation	Solutions, Partner validation	Reorder	*"Value proposition and Solution fit happen during value articulation to identify the problem, post which we ask for technology"*	MM
Value Identification	Business Value Articulation	Solutions	Reorder	*"Post business value articulation, need solutioning"*	DP-5
Value Identification	Business Value Articulation	Solutions	Change	*"There is a lot of hypothesis-based testing during Value Identification. This is a learning journey with low fidelity protype testing (paper prototype, mock UI screens to test market)"*	MM
Value Identification	Business Value Articulation	Partner validation	Addition	*"Partner validation critical - many partners have their own problems to solve"*	MM
Value Identification	Business Value Articulation	Pricing & ROI re-validation	Change	*"Budget and ROI are considered. We should also look at risk a project is carrying. The risk could be market (if market takes a different shift, you will stop the project) and technology (competition and is technology even tested in house? If not, then risk is higher)"*	MM
Value Identification	Co-creation alignment	Technology Considerations	Reorder	*"How business positions the product in market is important, post which technology comes into play"*	DP-5
Value Identification	Co-creation alignment	Technology Considerations	Change	*"Every 3 years, technology is changing and hence there is a need for adoption. The type of industry also matters (healthcare does not adopt tech as fast as say automobile or retail)"*	MM
Value Iteration	Co-create with partners	Product/Service marketing	Addition	*"Identify target customers for launches - leverage data and knowledge for customer partnership"*	DM-2
Value Iteration	Co-create with partners	Proof of Concepts / Rapid Prototype	Change	*"For POCs, technology validation is best, though at times there is a lot of wasted effort for example the spending we have had on Blockchain"*	DM-1
Value Iteration	Market Trials	Proof of Concepts / Rapid Prototype	Change	*"For Solution fit, we check if we make money? This is done by sketching/wireframes and showing to early adopters, determining how much are customers willing to pay. If this makes business sense, then move forward, then identify the technology. At this stage, we chase Research & Development team to get a prototype (called as Minimum Viable Product) and get this tested"*	MM
Value Realisation	Product/Service used by end customers	Product/Service marketing	Change	*"Identification of Distribution channels, Campaign management, Go-No go decision making, Supply chain management"*	MC-1
Value Realisation	Product/Service used by end customers	Measure value seen by the customer, ROI validation	Change	*"Measure the value seen by customer across both time and price. The ROI validation scales could be different for pilot* vs *scaled up model"*	DM-2
Value Realisation	Product/Service used by end customers	ROI validation	Change	*"Business model needs to be royalty based. What happens if business does not materialize in the longer term, leading us to determine how to leverage other parties. Also, we need to look at including sustenance part of realisation and how does it help in the growth"*	DP-5
Value Realisation	Product/Service used by end customers	Learnings and Adaptations to product/ service	Change	*"During realisation stage, we look for active usage in the market. This could lead to new feature or scope enhancements and the product evolves. Time is a key factor to realize value"*	MM
Value Realisation	Product/Service used by end customers	Learnings and Adaptations to product/ service	Change	*"Support is required, a life cycle by itself (customer feedback and support) and this could iterate"*	DP&E
Across the stages	Across	Across	Feedback loop	*"Post Learnings in iteration, we either Pivot (direction is right, but needs a different view), Persevere (continue) OR Stop (don't continue since it's a wasted effort)"*	MM
Across the stages	Across	Across	Feedback loop	*"Iteration loop - More like a start-up. 90-day meter to learn, process/ how-to based on budget, show prototypes, concepts and willingness of the customer to pay for it"*	MM
Value Iteration to Value Realisation	Across	Across	Feedback loop	*"Unless the problem is a 'burning need' - the POCs do not convert beyond for realisation"*	CIO
Value Iteration to Value Realisation	Across	Across	Feedback loop	*"Both stages – value iteration and value realisation – are iterative"*	MC-1

(*continued on next page*)

Table 5 (*continued*)

Value Co-creation Stage	Process Model	Step	Addition/ Change/ Reorder	Representative quote from interview	Respondent Designation
Pre-conditions	Governance, Trust	NA	Addition	*"How do the firm/supplier hold the IP to ensure that niche is with the firm/supplier"*	DP-5; MM; DP-4
Pre-conditions	Governance, Trust	NA	Addition	*"Look at IP protection (a) technology (b) Data. Does the contracting be different when supplier does not have the data. Hence a need for proper contract management"*	MC-2
Pre-conditions	Compliance & Regulatory	NA	Addition	*"Compliance/regulatory needs to be considered". "Who owns the IP? And is this more for firm or supplier driving the whole process"*	DP&E
Pre-conditions	Supply Chain Contract Management	NA	Addition	*"What could also be looked at is - what is the execution model (is it outcome/Fixed price based OR team to work through on POCs with customer team for innovation and later to do development). Hence look at contracting and governance model"*	TA

the type of technology that is used. The table also provides input in terms of the benefits that could be obtained due to the usage of digital technology in this process step of the VCC process with a few industry examples that came up during the interviews. For instance, in Value iteration stage and co-create with partners process model, one of the process steps is on Ideation/Brainstorming. Firms shall leverage transformation technologies such as Artificial Intelligence or Mind mapping tools or cloud-based collaboration tools for brainstorming ideas with their suppliers. This provides the firms and the suppliers with easy access to design thinking tools which helps in faster collaboration or idea generation leading to faster time to market and quality of the product.

The digital technology maturity could be either traditional, transitioning or transformative depending on the context of the engagement between the firm and supplier. We also observe that for every step in the process model across the Value Identification, Value Iteration and Value Realisation stages, digital technology could be leveraged by the firm and supplier when they co-create. The type of technology and the investment potential plays a key role in this adoption. The usage of digital technology suggested in this paper is thus an extension to the extant literature (Aslanova & Kulichkina, 2020; Boufim & Barka, 2021). The usage of the digital maturity curve in terms of Traditional, Transitioning and Transformative technologies as suggested in this paper is thus in line with the studies by O'Leary (2009), Ryan et al. (2020) and Kane et al., 2017. While the vehement response based on the interviews is that digital technology aids in faster realisation of the outcome, further research is required to have empirical evidence on the same. This is in line with the suggestion by Kane et al. (2017) considering the nascency of the digital technologies and their adoption being low.

5.2. Integrated framework and process model

Businesses today needs to continuously evolve, leveraging partners with the expertise to collaborate and co-create products that are meaningful to the end customer. Enabling the business in this process is digital technology, which provides benefits in terms of time to market, cost savings and quality. Digital technology is spreading at a rapid pace and newer technologies such as AI, IoT, GenAI, VR, and Smart Glasses are being used during VCC. Our findings suggest an approach to how businesses can use the 3 × 3 integrated framework to determine the cell in the matrix they are in and what they could do to enhance their reach to the end customer. Getting end customers as part of this journey is also very critical for businesses today as this helps validate the product/service much before it hits the market. End customers are thus part of the process and are seen as data providers assisting in the process of product performance, elevating the insights from the data and ensuring they are part of the digital transformation journey. End customers are also involved in bringing in their experiences to help validate ideas by businesses or set expectations in terms of what value means to them. Usage of social media is extensive, and it is no surprise that extensive literature of the 37 peer-reviewed articles focus on how social media is used for VCC. Social media is seen to enhance the reputation and knowledge about the company, stickiness to the brand and companies, and helping in the relationships leading to innovative products.

The 3 × 3 integrated framework brought out the fact that there remains much work to be done in terms of how business can iterate to get a testable product and then translate that to a measurable outcome, while leveraging digital technology. This is in line with the findings by Kane et al. (2017) who highlighted the adoption of digital technology being nascent. With the rapid innovations in digital technology in the past few years, the pace of realisation of value by leveraging these technologies during VCC in B2B companies is limited. The interviews helped us understand the gap that exists in literature today (Table 10; more in section 5.5), and how business continue to leverage newer digital technologies to co-create products, across the three VCC stages based on the digital maturity curve. This is in line with the findings by Boufim and Bark (2021) and Aslanova and Kulichkina (2020) who highlighted that digital technology helps drive in transformation of business and operational efficiencies.

But it is not just the framework alone that would help the academicians and practitioners. Given the complexity of B2B business compared to B2C and entire ecosystem play for VCC, it is even more critical to have a process model that would guide researchers and practitioners to leverage digital technology during VCC. Our findings in this study provide a process model from a VCC context (Fig. 6), discussing the three stages of VCC to a finer depth to provide insights to researchers and practitioners. The process model suggested is expanding on the extant research related to process models (Durugbo & Pawar, 2014; Grafmüller et al., 2018; Gronroos, 2011; Hein et al., 2019; Payne et al., 2009), but with specificity for the scope of this research.

Each process and step have a defined outcome and Table 10 highlighted how digital technology could play a vital role in enabling VCC. The benefits that businesses gain would be in terms of time to market, cost and quality. While proper identification of a gap or problem is a key need, as this would drive actions by the business in articulating the value and benefits arising from the product/service, it is also very critical to have a testable product/service quickly to ensure businesses can cut losses. Providing the platform for the firm and supplier to co-create meaningfully happens with a strong trust level amongst the parties involved in VCC, governance mechanisms, agreement on compliance and regulatory provisions and supply chain contract management.

The process model indicated that the process is iterative. Practitioners using the process model could for their specific needs go back

Table 6
Process Model – Value Identification.

Representative quotations	VCC Stage
"Company strategy is one level up of expert opinion? Consider 'market potential' for customer need" (DP&E) *"Need a business model (more than just governance) - how does it work/engagement model ('what is in it for me' for both parties) and why should customer do it? Why should we work together, more so in a connected world" (DP-5). "How will partner see value - what is the value pitch for partner in "should we co-create" (DP&E; DP-4)* *"Technology is accessible to anyone, but focus is required on business problem. How business positions the product in market is important, post which technology comes into play" (DP-5)* *"SWOT (Talent, technology and operating model) part of business value articulation since ability of the firm to deliver value needs to be considered for value positioning" (DP&E)* *"Value positioning and differentiation of product is critical" (DP-2)* *"Market research is typically given to third-party and hence could be misleading as they answer the questions we ask. For understanding Value in terms of money, it is thus important to understand VOC (voice of customer), talking to personas such as lab personnel, middle management and decision maker" (MM)* *"Business Value Articulation: Cost benefit analysis, Feasibility studies" (MC-1)* *"Pricing and ROI considerations come during business value articulation" (DM-2)* *"Value proposition and Solution fit happen during value articulation to identify the problem, post which we ask for technology" (MM)* *"Post business value articulation, need solutioning. Does not technology come into play at this point?" (DP-5)* *"Solutioning happens during value identification, where we identify the potential bottlenecks and how do we get them resolved" (DP&E)* *"In case of large gap - business may split into smaller pieces to realize value (MVP) - 1 small pilot to give more confidence" (DP&E)* *"Interlock with partners required during value identification" (DM-2)* *"We first look at gap identification and what is meaningful to both parties, then build a solution and iterate" (CMO)* *"Ideation/Brainstorming shall happen in Value identification" (DM-2)* *"In Identify gap, we need to look at parameters involving in-house budget and funding needs (leading to funding-based partner decision for pilot and scale)" (DP-3)* *"Apart from identifying gaps, also look at unutilized opportunity" (CTO)* *"There is a lot of hypothesis-based testing during Value Identification. This is a learning journey with low fidelity protype testing (paper prototype, mock UI screens to test market)" (MM)* *"Partner validation critical - many partners have their own problems to solve" (MM)* *"To measure success for value measurement: We talk to 10 customers and if 70 % of our customers say this is a problem, then this becomes a checkpoint" (MM)* *"Every 3 years, technology is changing and hence there is a need for adoption. The type of industry also matters (healthcare does not adopt tech as fast as say automobile or retail)" (MM)* *"Budget and ROI are considered. We should also look at risk a project is carrying. The risk could be market (if market takes a different shift, you will stop the project) and technology (competition and is technology even tested in house? If not, then risk is higher)" (MM)* *"Post Learnings in iteration, we either Pivot (direction is right, but needs a different view), Persevere (continue) OR Stop (don't continue since it's a wasted effort)" (MM)*	Value Identification

and revisit a particular step in a stage or go back and re-start the entire process again. Considering that VCC outcomes can only be measured by the value that the end customer perceives, it is only natural for the

Table 7
Process Model – Value Iteration.

Representative quotations	VCC Stage
"Unless the problem is a 'burning need' - the POCs do not convert beyond for realisation. There is intent at the top to use latest tech, but at the ground level there is resistance (want to maintain status quo)" (CIO) *"Identify target customers for launches - leverage data and knowledge for customer partnership" (DM-2)* *"Driver is from business and who holds the budget. ROI validation could be very different with various stakeholders (strategy of both organization needs to match). Hence sponsor identification and strategy teams have to work together with governance layer" (CDO)* *"For POCs, technology validation is best, though at times there is a lot of wasted effort for example the spending we have had on Blockchain" (DM-1)* *"For Solution fit, we check if we make money? This is done by sketching/ wireframes and showing to early adopters, determining how much are customers willing to pay. If this makes business sense, then move forward, then identify the technology. At this stage, we chase Research & Development team to get a prototype (called as Minimum Viable Product) and get this tested" (MM)* *"Both stages – value iteration and value realisation – are iterative" (MC-1)* *"Iteration loop - More like a start-up. 90-day meter to learn, process/ how-to based on budget, show prototypes, concepts and willingness of the customer to pay for it" (MM)*	Value Iteration

Table 8
Process Model – Value Realisation.

Representative quotations	VCC Stage
"During realisation stage, we look for active usage in the market. This could lead to new feature or scope enhancements and the product evolves. Time is a key factor to realize value" (MM). *"Identification of Distribution channels, Campaign management, Go-No go decision making, Supply chain management" (MC-1).* *"Measure the value seen by customer across both time and price. The ROI validation scales could be different for pilot* vs *scaled up model" (DM-2).* *"Support is required, a life cycle by itself (customer feedback and support) and this could iterate". "Should the process also have marketing/brand as part of this. Tech plays a big role being able to spot opportunity" (DP&E).* *"Business model needs to be royalty based. What happens if business does not materialize in the longer term, leading us to determine how to leverage other parties. Also, we need to look at including sustenance part of realisation and how does it help in the growth" (DP-5).*	Value Realisation

Table 9
Process Model – Preconditions for VCC.

Representative quotations	VCC Stage
"How do the firm/supplier hold the IP to ensure that niche is with the firm/supplier" (DP-5; MM; DP-4) *"Look at IP protection (a) technology (b) Data. Does the contracting be different when supplier does not have the data. Hence a need for proper contract management" (MC-2)* *"Compliance/regulatory needs to be considered". "Who owns the IP? And is this more for firm or supplier driving the whole process" (DP&E)* *"What could also be looked at is - what is the execution model (is it outcome/Fixed price based OR team to work through on POCs with customer team for innovation and later to do development). Hence look at contracting and governance model" (TA)*	Pre-conditions

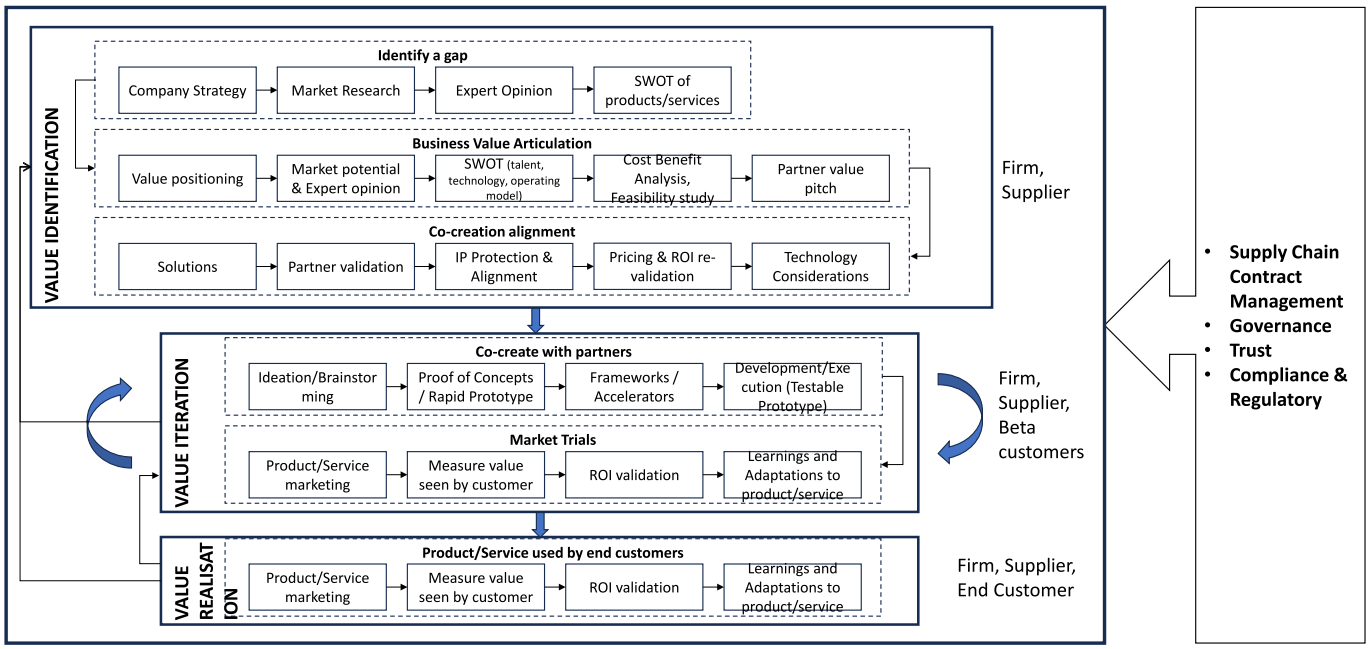

Fig. 6. Process model for co-creation with usage of digital technology.

processes/steps to be iterative in nature. The integration of the 3 × 3 integrated framework with the process model provides opportunities for researchers and practitioners to take a more holistic view of how digital technology could be leveraged better during VCC in B2B environment. A faster feedback loop (due to usage of technology in each of the process steps) provides opportunities for business to save time, cost and take right product/service to the market.

5.3. Research implications

From a research perspective, we have three main contributions. First, the 3 × 3 integrated framework (Fig. 5) provides a good foundation for research performed until now. This shows the gap seen in terms of more research being needed in Value Iteration and Value Realisation for transitioning and transformative technologies. The extension of the research by Boufim and Barka (2021) and Aslanova and Kulichkina (2020) helps us understand the nuances of the interaction between VCC and digital technologies. Second, the study thus extends work done by Grafmüller et al. (2018) covering the detailed aspects in terms of information system design during the process of VCC in a B2B environment while leveraging digital technology in the process. Given that B2B is complex compared to B2C, the process model shall help researchers establish the usage of digital technology in a succinct and easy way. As indicated in the process model, there are pre-conditions that drive VCC in B2B context. Trust has been discussed in detail by Sundström et al. (2021) and Falkenreck and Wagner (2022). Based on our findings, it is critical for firms and suppliers to also consider aspects related to supply chain contract management, Governance and compliance and regulatory aspects to have a long-term successful partnership. Third, the integration of the 3 × 3 integrated framework with process model helps provide a holistic view on digital technology adoption during the process of VCC in B2B environment. Determining the right digital technology to be used in each of the process step would go a long way in terms of understanding the rationale and reason for adoption from a researcher's perspective.

5.4. Managerial implications

From a managerial perspective, there are four main contributions. First, the usage of digital technology to aid in the VCC stages across the digital maturity curve (Fig. 6 and Table 3) shall be a good starting point. This would aid the managers to find additional ways to co-create to gain faster time to market/speed, ensure quality and overall reduce the cost. This would also help the firms to realize a higher market share in the longer run by leveraging digital technology during VCC. While value identification is easier, firms are finding value iteration and measurement/realisation difficult due to the complexities of the ecosystem in a B2B context. Second, we find that VCC has a key role to play in reducing the failures in adopting transformative technologies. The buyer and supplier firms need to keep themselves abreast of the rapidly changing technology. Digital technology plays a key role in being able to spot opportunities and aids the process of VCC in the dyadic relationship between buyer and supplier. Third, the study brought out a validated process model (Fig. 6), which if used during the process of VCC, aids in realisation of value at a higher speed, gain efficiencies, cost differential, sustainable development and acquire more customers, thus providing gains for business. Fourth, the study has brought a ready reckoner on the type of digital technology that a manager of a firm or supplier could use for a particular process step as illustrated in Table 10. The table shall also help the managers to determine the benefit they would get due to the usage of digital technology in the VCC process, with some examples helping them in their journey at their organization. In a competitive world with rapid growth of digital technologies, the buyer and supplier do not have an option but to leverage technology and grow together.

Table 10
Leveraging digital technology during co-creation stages for the process model.

Value Co-creation stage	Process Model	Step	Usage of Technology	Digital Technology Maturity	Type of Technology	Benefit with usage of technology	Examples
Value Identification	Identify a gap	Company Strategy	Yes	Transformative Technologies	Big Data, Artificial Intelligence (AI), Machine Learning (ML), Cloud Computing	Time to market/ Speed, Cost savings, Quality	Usage of technologies to get meaningful insights based on data collected, automate tasks, predict future trends and identify insights - helping in the overall formulation of company strategy
Value Identification	Identify a gap	Market Research	Yes	Transformative Technologies	Online surveys and polls, CRM, Social Media, web crawlers, SEO traffic, consumer intelligence and research tools (NielsenIQ, SurveyMonkey etc), Market Analysis tools (QuestionPro etc), Data analytics s/w, Data recorder S/W, Competitive intelligence tools,	Time to market/ Speed, Cost savings	Unilever using technologies to co-create with its dealers
Value Identification	Identify a gap	Expert Opinion	Likely	Traditional and Transformative Technologies	Video conferencing/transcription software, Artificial Intelligence (AI); Machine Learning (ML); Natural Language Processing (NLP)	Time to market/ Speed, Cost savings	Gartner uses technologies such as video conferencing/transcription software and AI (for focus groups, expert interviews) to collect feedback from experts. Forrester uses ML to predict future trends and outcomes in industries. IDC uses NLP to analyze feedback and identify key insights. B2B firms use Gartner and Forrester reports to draw their product strategy roadmap
Value Identification	Identify a gap	SWOT of products/ services	Yes	Transformative Technologies	Online surveys and polls, Collaboration tools, Mind Mapping software; Artificial Intelligence (AI); Machine Learning (ML); Natural Language Processing (NLP)	Time to market/ Speed, Cost savings, Quality	Google. Microsoft, Amazon and various such Firms use multiple technologies to conduct SWOT analysis - based on both structured and unstructured data
Value Identification	Business Value Articulation	Value Positioning	Yes	Transformative Technologies	Big Data, Artificial Intelligence (AI), Machine Learning (ML), Cloud Computing	Time to market/ Speed, Cost savings, Quality	Amazon uses AI and ML that provide predictive models to forecast future customer needs and preferences. IT Firms leverage technologies to build platforms which help in the positioning for co-creation to firms
Value Identification	Business Value Articulation	Market Potential & Expert Opinion	Likely	Transformative Technologies	Video conferencing/transcription software	Time to market/ Speed, Cost savings	Firms use technologies such as video conferencing/ transcription software for focus groups and expert interviews to gauge the investment spend
Value Identification	Business Value Articulation	SWOT (talent, technology, operating model)	Yes	Transformative Technologies	Online surveys and polls, Collaboration tools, Mind Mapping software; Artificial Intelligence (AI); Machine Learning (ML); Natural Language Processing (NLP)	Time to market/ Speed, Cost savings, Quality	Google. Microsoft, Amazon and various such Firms use multiple technologies to conduct SWOT analysis - based on both structured and unstructured data
Value Identification	Business Value Articulation	Cost Benefit Analysis, Feasibility study	Yes	Traditional or Transitioning Technologies	CRM and Platform; Big Data, Social Media, Cloud Computing	Time to market/ Speed, Cost savings	Usage of CRM and platforms within or inter-Firms to enable calculation of ROIs. Firms leverage technologies to get insights from data, identify insights, forecast future size of market, avoid upfront investments
Value Identification	Business Value Articulation	Partner value pitch	Yes	Transformative Technologies	CRM, Social Media	Time to market/ Speed, Cost savings, Quality	Identifying "what is in for the partner" rationale
Value Identification	Co-creation Alignment	Solutions	Yes	Transformative Technologies	Data Analytics, Artificial Intelligence (AI), Machine Learning (ML), Internet of Things (IoT)	Time to market/ Speed, Cost savings, Quality	Firms leverage technology (Data Analytics, AI, IoT) to collect, analyze data, identify patterns and insights to determine the solutions to close the gap. ML can be used to develop predictive models to forecast future trends and outcomes.
Value Identification	Co-creation Alignment	Partner validation	Yes	Transitioning and Transformative Technologies	CRM, Artificial Intelligence (AI); Machine Learning (ML); Big Data; Social Media	Time to market/ Speed, Cost savings, Quality	Usage of technologies by firms to validate partner capabilities for co-creation
Value Identification	Co-creation Alignment	IP Protection & Alignment	Yes	Transitioning and Transformative Technologies	Data encryption, data watermarking, access control, digital rights management, blockchain, Artificial Intelligence (AI), Machine Learning (ML)	Time to market/ Speed, Cost savings, Quality	Firms such as Netflix, Apple or Microsoft use technologies to protect their Intellectual property using DRM technologies to prevent users from copying, distributing of modifying the content

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Table 10 (*continued*)

Value Co-creation stage	Process Model	Step	Usage of Technology	Digital Technology Maturity	Type of Technology	Benefit with usage of technology	Examples
Value Identification	Co-creation Alignment	Pricing & ROI considerations	Yes	Transformative Technologies	Big Data, Artificial Intelligence (AI), Machine Learning (ML), Cloud Computing	Time to market/ Speed, Cost savings, Quality	Amazon uses AI and ML to determine forecast future demand and prices; Google uses AI, ML and Cloud computing to determine the future advertising demand and prices. Insights are also driven through data collected
Value Identification	Co-creation Alignment	Technology considerations	Yes	Transitioning and Transformative Technologies	CRM, Artificial Intelligence (AI); Machine Learning (ML); Big Data; Social Media; Collaboration tools	Quality	Firms leverage technologies to determine the technology to be used for co-creation
Value Iteration	Co-create with partners	Ideation/ Brainstorming	Yes	Transformative Technologies	Artificial Intelligence (AI); Mind mapping tools (Whiteboard, flowcharting tools etc.), cloud-based collaboration tools, Design thinking tools, conferencing apps (zoom etc)	Time to market/ Speed, Quality	Firms leverage technologies/software for idea generation, trend analysis, brainstorming assistance, process optimization; record and link together ideas
Value Iteration	Co-create with partners	Proof of Concepts / Rapid Prototype					
Value Iteration	Co-create with partners	Frameworks/ Accelerators					
Value Iteration	Co-create with partners	Development/ Execution (Testable Prototype)	Yes	Transformative Technologies	3D printing, Laser cutting, Containerization, Orchestration, CI/CD, Design thinking tools (Figma), modern Rapid application Development (RAD) software	Time to market/ speed, Quality, Cost Reduction	Firms leverage technologies for building a testable prototype
Value Iteration	Market Trials	Product/Service marketing	Yes	Traditional or Transitioning Technologies	Content Management Systems (CMS), Customer Relationship Management (CRM), Marketing automation software, Search engine optimization (SEO), Pay-per-click advertising (PPC), Social Media; Artificial Intelligence (AI), Machine Learning (ML), Augmented and Virtual Reality (AR/VR), Blockchain	Time to market/ speed, Quality, Cost Reduction	Firms leverage multiple digital channels for campaign management, leveraging data to identify Beta customers and do targeted marketing for the prototype to elicit feedback. Companies such as Amazon use AI to personalize message and to determine effectiveness of their marketing campaigns
Value Iteration	Market Trials	Measure value seen by customer	Yes	Transformative Technologies	Artificial Intelligence (AI), Virtual Reality (VR), Augmented Reality (AR), Feedback (micro survey using Google forms etc., Feedback software)	Access to new markets, Time to market/speed, Quality, Cost Reduction	Customers of Starbucks use interactive experience to learn more about coffee and provide feedback. Starbucks allows customers to customize their drinks - thereby creating new products. Customers of Nike try the new products and provide feedback, based on which Nike makes changes before launch of their product
Value Iteration	Market Trials	ROI validation	Yes	Traditional or Transitioning Technologies	CRM and Platform	Time to market/ Speed, Cost savings	Usage of CRM and platforms within or inter-Firms to enable calculation of ROIs
Value Iteration	Market Trials	Learnings and Adaptations to product/service	Yes	Transformative Technologies	Online surveys and polls, Feedback (micro survey using Google forms etc., Feedback software), Cloud based platforms/tools	Time to market/ Speed, Cost savings, Quality	Collect feedback from the beta customers to make changes to the product/service
Value Realisation	Product/Service used by end customers	Product/Service marketing	Yes	Traditional or Transitioning Technologies	Content Management Systems (CMS), Customer Relationship Management (CRM), Marketing automation software, Search engine optimization (SEO), Pay-per-click advertising (PPC), Social Media; Artificial Intelligence (AI), Machine Learning (ML), Augmented and Virtual Reality (AR/VR), Blockchain	Time to market/ speed, Quality, Cost Reduction	Firms leverage multiple digital channels for campaign management, leveraging data to identify Beta customers and do targeted marketing for current customer and to gain traction with new customers. Companies such as Amazon use AI to personalize message and to determine effectiveness of their marketing campaigns

(*continued on next page*)

Table 10 (*continued*)

Value Co-creation stage	Process Model	Step	Usage of Technology	Digital Technology Maturity	Type of Technology	Benefit with usage of technology	Examples
Value Realisation	Product/Service used by end customers	Measure value seen by customer	Yes	Transformative Technologies	Internet of Things, feedback (micro survey using Google forms etc., Feedback software)	Cost savings, Quality, New products	IoT devices help track inventory levels in real time or monitor customer behaviour in store or provide inputs for calculating Remaining Useful Life (RUL) - helping improve the supply chain management, customer experience, downtime of machines
Value Realisation	Product/Service used by end customers	ROI validation	Yes	Traditional or Transitioning Technologies	CRM and Platform Artificial Intelligence (AI) - multi-intent conversations through GenAI, Social Media, Data Analytics software, Digital immune technologies (ensures reliability), feedback (micro survey using Google forms etc., Feedback software), Cloud based platforms/tools,	Time to market/ Speed, Cost savings	Usage of CRM and platforms within or inter-Firms to enable calculation of ROIs
Value Realisation	Product/Service used by end customers	Learnings and Adaptations to product/service	Yes	Transformative Technologies		Time to market/ Speed, Cost savings, Quality	Use of AI and Social Media to analyze feedback from customers by Nike/Starbucks - giving faster feedback and leading to improvements. Usage of data analytics software for feedback (big data) to improve the product/service

5.5. Limitations and future research avenues

Although this study provides valuable contribution to the B2B literature in the domain of usage of digital technology in VCC stages, the study has some limitations. As the literature has been carried out on a specific date, coverage of all existing peer-reviewed papers in this field is not performed. The clustering of the papers into the three VCC stages, leading to the 3 × 3 integrated framework, is based on author's thorough reading of the papers and is not grounded. But we believe that integrated framework is the first step towards a thorough understanding of the VCC stages and the extant literature. There are trends in usage of AR/VR for enhancing customer experience in retail stores, identifying potential failures and providing alerts in an oil and gas setup. B2B companies continue to have opportunities to leverage newer digital technologies such as Big Data, AI, Industry 4.0, Cloud Computing, AR/VR, etc. in multiple perspectives across each domain. This is even more important in a platform environment that service industry is moving towards with an ecosystem consisting of multiple actors. The studies should also focus on how co-destruction is likely with latest technological advancements, to ensure that companies take the right precautions in setting up their environment currently. Future research can also extend learnings from innovations in customer experience in the retail and travel industry with usage of AR/VR. The framework developed based on the semi-structured interviews can help research to appreciate the practical usage of digital technology in the industries. The interviewees indicated the additional uses of digital technology in the VCC stages, as illustrated in Fig. 7, which provides direction for future research. Thus, the findings unlock possibilities for future research.

We expect variations based on the firms involved in VCC in the process model steps of co-creation alignment and co-create with partners. Further research leveraging case studies is expected to refine these steps and typologies built accordingly. This would then act as a ready reckoner for academics to synthesise the maturity of the field and for practitioners to leverage the insights. The process model has been arrived at based on the interviews conducted across the wide spectrum of roles/titles. One of the limitations would be that 56 % of the interviewees are from Information Technology sector and hence there is scope for further broadening the study by conducting interviews across other sectors. This would help refine the process model and the associated use of digital technology to aid in the VCC stages.

Today's digital world adopts the usage of AI for various financial and non-financial transactions across the industries. Usage of chatbots, self-service kiosks, service robots, and automated transactions all help the end customer, thereby providing opportunities for B2B companies to collaborate and co-create using the latest technologies. While there is some research done in this area, extending this to the value iteration and value realisation stages of VCC would go a long way in adoption of transformative technologies for co-creation. The customer's perception, trust, social influence, hedonic motivations and prior experience influence the usage of AI in tourism and hospitality industry (Solakis, Katsoni, Mahmoud, & Grigoriou, 2022). This finding is in line with the call for further research in AI and robots in VCC (Kaartemo & Helkkula, 2018).

Amongst the reviewed papers, it is observed that the amount of quantitative research papers is low. Much of the work on VCC in B2B context remains at a conceptual level and hence there is a need for answering some of the questions such as how conceptual learning can be validated through quantitative studies on the digital technology usage in VCC stages; will some of the quantitative studies lead to a changed contextual understanding for the VCC stages in B2B. In a multi-dyadic, platform ecosystem, the various complexities in such an environment shall lead to changes in theoretical assumptions.

While S–D logic remains prominent, the fact that more than 50 % of the papers do not mention a theoretical lens is a gap that needs to get addressed. What is likely to help for adoption of digital technology in the VCC stages of value iteration and value realisation is the perspective of

Digital Technology Maturity	Value Identification	Value Iteration	Value Realization
Transformative Technologies	1. **GenAI**: personalized medical treatments, new drug discovery or scientific discoveries, virtual worlds to explore different cultures, personalized and engaging experience for students of all ages 2. Leveraging **blockchain**, producers in India need to establish the source of cotton as this is a pre-requisite for export of Organic clothes to European dealers	1. **GenAI**: Usage of LLM for implementation of solutions for contract creation, healthcare applications to suggest treatments and provide patients with information and support 2. **IOT+ML**: Reduced legal and financial liabilities and help recover batteries with co-creation between Electric Vehicle companies, battery manufacturers and solutions by 3rd party companies which help in calculating Remaining Useful Life (RUL) 3. **Blockchain** technology used in P&G's procure-to-pay to cover the actual terms of the contract. AI could also be used further for order fulfilment	1. **Smart Sensors:** Waggeryd installed the ABB Ability Condition Monitoring solution digitalizing their entire critical line. The sensors effortlessly collect and transmit data to the ABB Ability Condition Monitoring system that analyzes the information to provide key operational parameters of the equipment and give early warning about potential trouble that may be developing; allowing the mill managers to take a well-informed approach about what motors to replace and with what priorities. 2. **Smart Meters:** Major electricity and water providers in the Middle East collaborated with Honeywell to provide "Smart electricity meters" helping the providers to charge the end-customer for their consumption. Using Honeywell's technology, two-way communication can facilitate on-demand meter reading and transform grid management. 3. **GenAI**: Usage of LLMs for customer service chatbots, personalized learning experiences for students in education sector, creative writing tools, personalized travel recommendations, travel planning in travel sector, product recommendations and marketing campaigns in retail sector 4. **IOT+ML+Big Data**: Usage of Industrial IOT along with Machine Learning on Big data for determining the Remaining Useful Life (RUL) of turbines in Ship 5. **IOT based process automation** helped an Australian farmer to use 20% less water due to new IoT-based irrigation system supplied by Lindsay (US based manufacturer) 6. **Remote Asset Monitoring** by Coke-Cola for its PET and can-bottling lines using **IoT**, partnering with its co-creation partners 7. **AR device** to reduce mishandling of bags at Changi Airport. The AR device has ensured that baggage handlers cannot go wrong and provides the traceability for loading through digitization
Transitioning Technologies	1. **AI** can enhance collaboration between airlines, airports and retailers by providing a platform to advertise products for airline customers, with airports providing the fulfilment		1. **Customer Relationship Management (CRM)** of Salesforce redesigned to embrace the native iOS platform of Apple 2. Cigna partnered with Pega with implementation of a **core health platform** enabled new models of payer and provider collaboration, support innovative consumer capabilities and increase the business agility 3. Apple and Salesforce have developed **first Salesforce Mobile SDK** (software development kit) which is optimized for Swift, Apple's programming language 4. **Virtual accounting platform** used by ICICI Bank using API integration removing accounting errors (reconciliation issues) 5. Anti-counterfeiting with secure **QR codes** for Exxon Mobile oil leveraging Scantrust 6. Integration of digital identity during Covid for providing health passports in India leveraging **QR codes** and **Blockchain** technologies
Traditional Technologies			

Value Co-creation Stages

Fig. 7. Integrated framework of digital technology maturity and VCC stages – areas of potential research.

conceptual frameworks and theoretical lens that could be leveraged from operations and service management literature. The future focus in this specific area would be on how theory can aid the technological advancements seen in the industrial industry, and what are the implications from a theoretical perspective on the VCC stages based on multiple stakeholders involved in the VCC ecosystem, amongst others.

6. Conclusion

With the rapid advancement of technologies, there is a need for determining the usage of digital technology in VCC in B2B environment involving buyer and supplier (dyadic relationship). By reviewing the literature in detail and conducting interviews, this study has answered the two research questions raised at the beginning of the paper. To validate RQ1, the review of the 37 peer-reviewed papers showcases the adoption and the positive impact of digital technology in VCC. Digital technology has enabled VCC across multiple industries/business and across the entire life cycle of VCC. An integrated 3 × 3 framework is developed illustrating the usage of various technologies on the VCC stages. Associated with this, based on the 25 semi-structured interviews, the validation of the framework along with potential areas of research is a way forward for theoretical research. The recent advancements in digital technology and how these could be leveraged in plugging the key gaps in current literature, additionally answers RQ1. The process model, an outcome of the semi-structured interviews, forms a basis for usage of digital technology during each step of VCC (answering RQ2). The answers can be ready reckoners for researchers to assess the intellectual structure of the field and for practitioners to choose the right methodology and digital technology.

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CRediT authorship contribution statement

Rajesh Shankaranarayana: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Gopalakrishnan Narayanamurthy:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Conceptualization. **Shalini Ramaswamy:** Writing – review & editing, Validation, Methodology, Formal analysis, Data curation. **Roger Moser:** Writing – review & editing, Validation, Supervision, Methodology.

Declaration of competing interest

None.

Appendix A. Supplementary data

Supplementary file with Appendix 1-4 can be found online at https://doi.org/10.1016/j.indmarman.2024.12.007.

Data availability

The authors do not have permission to share data.

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